

The Morphology/Syntax Interface

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March 2026

Pre-publication version

In press at Cambridge University Press

Elements in Generative Syntax

Acknowledgments

This manuscript was a true team effort, written over several years of both virtual collaboration and in-person writing retreats. Thanks are due to many individuals who offered comments, helpful discussion, or other support along the way, who all helped make this project successful. First of all, we would like to thank Vrinda Chidambaram, Vera Gribanova, Heather Newell, and Sylvia Schreiner, who participated in our writing group and writing retreats and offered insightful comments and moral support throughout the process of writing this Element. Thanks also to our editor, Bob Freidin, who waited very patiently (and encouragingly) for us to finish the manuscript, and who helped us in clarifying and focusing our presentation. For illuminating discussions we thank Pavel Caha, Matthew Hewett, Nicholas Rolle, Dan Siddiqi, and Coppe van Urk. For detailed comments on the manuscript we thank Jonathan Bobaljik, Neil Myler, and our anonymous reviewers—our Element benefited enormously from your feedback. For top-notch assistance with various editing tasks, we thank Maksymilian Dabkowski and Alexander Elias. The other authors would like to especially thank Laura Kalin, for shepherding us through this project from its very inception, and without whom it would never have been finished.

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1 Introduction

Linguistic expressions vary in the complexity of their form and meaning, as illustrated across the set in (1):

- (1) a. fund
- b. nonrefundable
- c. not able to get money back

Generative theorists generally agree that a linguistic expression like that in (1a)—whose surface string cannot be broken down any further into meaningful subparts—is stored in some fashion as an arbitrary form-meaning correspondence.¹ A linguistic expression of phrasal size, as in (1c), is also uncontroversial: it is the purview of the generative system known as THE SYNTAX. Much more contentious is how to treat a linguistic expression like that in (1b), a complex WORD, which has in common with (1c) that it is made up of multiple meaningful subparts, but in a more tightly-packed configuration, potentially on par in certain respects with (1a).

Words—and all that goes on inside words, including how they’re built and what determines their form—constitute the empirical domain of (lower case m) MORPHOLOGY. Larger linguistic expressions, of phrasal and clausal size, constitute the empirical domain of (lower case s) SYNTAX. A major question for any generative theory of grammar is, what is the relationship between words (and what goes on inside them) and larger linguistic expressions (and what goes on inside them)? This relationship, known as the MORPHOLOGY/SYNTAX INTERFACE, is a richly-discussed and hotly-debated area of linguistic theorizing, deeply rooted in the history of the field. Foremost in understanding this interface is determining whether there is a generative system specifically dedicated to some or all aspects of morphology, something we could refer to as THE MORPHOLOGY, as distinct from the Syntax.

This Element tackles major empirical phenomena and theoretical questions that arise at the morphology/syntax interface, aiming to give readers both an aerial view of the theoretical landscape as well as the concrete tools needed to engage with this active and exciting research area. We begin by briefly motivating the interconnectedness of syntax and morphology, §1.1, introducing the Y model, §1.2, and laying out a roadmap, §1.3.

¹Though even a seemingly simple expression like that in (1a) still has a number of complexities: *fund* could be a noun or a verb, *fund* could be analyzed as bearing (e.g.) null number or tense morphology, etc. Nevertheless, all scholars would agree that *fund* belongs to some arbitrary, stored correspondence with a meaning, be that in its nominal form, in its verbal form, as an uncategorized root, or as a set of abstract features.

1.1 Morphology and syntax

There is abundant evidence that morphology and syntax are, in some way, interconnected. To start with an obvious example, DERIVATIONAL MORPHOLOGY determines the syntactic category of a word. For example, the morphologically-related words *jewel* and *bejewel* have different syntactic distributions attributable to the presence of *be-*, with *jewel* able to serve as the head of a noun phrase, and *bejewel* able to serve as the head of a verb phrase.

The relationship between morphology and syntax is also evident with respect to INFLECTIONAL MORPHOLOGY, traditionally subsuming the marking of tense, aspect, and mood, as well as case, person, number, and gender (a.o.). Compare the bolded words across (2a) and (2b), as well as the italicized words:²

- (2) Lithuanian (East Baltic; Milena Šereikaitė p.c.)
- a. **Motin-a** raš-o *laišk-a*
mother-NOM.F.SG write-PRS.3 letter-ACC.SG
‘The mother is writing the letter.’ (active)
- b. *Laišk-as* yra **motin-os**
letter-NOM.M.SG be-PRS.3 mother-GEN.F.SG
rašo-m-as
write-PASS.PTCP-NOM.M.SG
‘The letter is being written by the mother.’ (passive)

The syntactic context of each noun in (2) correlates with the noun’s morphological form. As the subject of the active clause in (2a), ‘mother’ bears a suffixal nominative case marker (also encoding gender and number), *-a*. When demoted in the passive in (2b), ‘mother’ bears a suffixal genitive case marker (again encoding gender and number), *-os*. Similar observations hold for ‘letter’ (italicized) across the two examples, as the object of the active in (2a) and the subject of the passive in (2b). The morphological form of the verb and auxiliary, too, reflect syntactic context, displaying agreement with a nominal expressed independently in the larger linguistic expression.

As a yet deeper connection between morphology and syntax, the relationships among subparts of a word often appear to be mirrored in the relationships among subparts of a clause, known as Mirror Principle effects (Baker, 1985). As observed by Bybee 1985 and Julien 2002 (among others), many elements

²Glosses in this Element follow the Leipzig Glossing Rules, with the following additions: HAB = habitual, THM = theme, DTR = ditransitive, LNK = linker, STV = stative, VAR = stem variation, PFX = prefix, REP = repetitive. Where necessary, glosses from other sources have been adapted to the Leipzig conventions. Examples without an accompanying citation originate with the authors of this Element.

follow an invariant relative order/hierarchy whether free-standing (as independent words) or affixal (inside a word). For example, observe the order of tense, aspect, and mood first in Yorùbá, (3a), where these are expressed with independent words, and next in Sabanê, (3b), where they are expressed with affixes inside the verb word. (Examples from Cinque 2014; verb root bolded.)

- (3) a. Yorùbá (Niger-Congo, West Africa; Oladiipo Ajiboye, p.c. to Cinque)
 Ñjé Adé yòò máa **wá** ní ìròlẹ́ [C T Asp **V**]
 Q Ade FUT HAB come in evening
 ‘Will Ade be coming in the evenings?’
- b. Sabanê (Nambikwaran, Brazil; de Araujo 2004, p. 206)
 Uli **ay**-i-say-al-a [V-Asp-T-C]
 2SBJ go-V-PROG-PRES-Q
 ‘Are you leaving?’

In both languages, the order is as follows: aspect is closest to the verb root (HAB, PROG above), tense is further out (FUT, PRES above), and mood is the most peripheral, i.e., farthest from the verb root (seen with the question marker Q above). Limited other orders are possible cross-linguistically as well, overwhelmingly respecting this basic hierarchy. While not exceptionless, such parallels between morphological ordering and syntactic ordering (and semantic scope) abound; see, e.g., Speas 1991; Cinque 2005; Alexiadou et al. 2007.

There must be *some* kind of relationship between syntax and morphology. It is this relationship we will investigate in this Element.

1.2 The Y model

Viewing linguistic expressions as pairings of form and meaning, a generative model of grammar must minimally include interfaces with the phonological component (Phonological Form; PF) and the conceptual component (Logical Form; LF), as argued in early Minimalist work (Chomsky, 1993, 1995b). Specifically, the architecture of the grammar in Minimalism has the shape of an inverted Y, as in Figure 1 (conceptually originating in Chomsky and Lasnik 1977). The lexicon provides the basic building blocks for the syntactic computation, which produces a syntactic representation. The syntactic representation at the point of spell-out provides the input to PF and LF.

This Element considers the interface of morphology with a Minimalist-style, derivational syntactic framework.³ The circled numbers in Figure 1

³Generative syntactic theories can, alternatively, be non-derivational, called “representational” or “constraint-based”; these include Lexical Functional Grammar (LFG; Bresnan

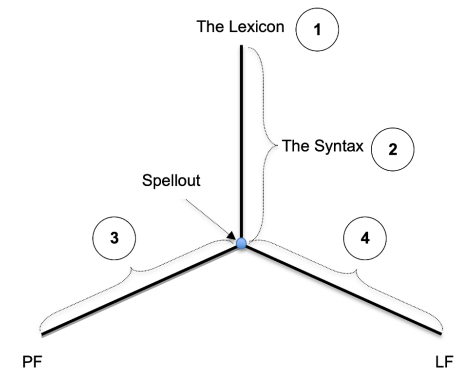


Figure 1: The Y-model, including possible points of contact with morphology

show where in the Y model morphological phenomena could be accounted for: pre-syntactically (point of contact 1), as part of the syntactic computational system (point of contact 2), on the branch to PF (point of contact 3), and/or on the branch to LF (point of contact 4). An additional possibility, not represented in Figure 1, is that morphology is accounted for outside of the Y.

1.3 Roadmap to the Element

In this Element, we ask: what are the prominent theoretical approaches to the study of the relationship between morphology and syntax? What are the linguistic phenomena that are most relevant at the morphology/syntax interface, and how are these handled in different types of theories? The theoretical landscape is the topic of §2 of the Element. §3 addresses questions about wordhood and word-formation, central to the very division between morphology and syntax in the first place. §4 contends with the (at least apparent) abundance of imperfect mappings in the relationship between meaning/function and form. We conclude in §5 by highlighting the importance of taking an aerial view.

We would like to note up front that we, the co-authors of this Element, adopt (in our own work) some or all aspects of the approach to the morphology/syntax interface known as Distributed Morphology (DM).⁴ Nevertheless, our goal in this Element is to accurately represent a wide variety of approaches.

1982) and Head-Driven Phrase-Structure Grammar (HPSG; Pollard and Sag 1994), as well as the development of Minimalism by Brody (2000, et seq.).

⁴For origins, overviews, and key features of the framework of DM, see Noyer 1992; Halle and Marantz 1993, 1994; Harley and Noyer 1999; Embick 2010; Bobaljik 2017, and the collections Matushansky and Marantz 2013 and Alexiadou et al. To appear.

2 Dimensions of the interface

In this section, we aim to elucidate the core properties of, and crucial differences among, generative theories of the morphology/syntax interface. In §2.1 we introduce the key dimensions of variation across theories, along with several fundamental concepts that are essential to the Element. The remainder of this section is organized around one of these dimensions: §2.2 discusses LEXICALIST approaches, which are those that posit a narrow interface between morphology and syntax, at the level of the word; §2.3 turns to the complement set, which we term NON-LEXICALIST approaches. §2.4 concludes the section with a summary table representing major differentiations across approaches.

2.1 Overarching concepts and dimensions of variation

Theories of the morphology/syntax interface vary along four major dimensions: (i) do words constitute the interface between morphology and syntax? (§2.1.1), (ii) how is the morphological computational system situated with respect to the syntactic computational system? (§2.1.2); (iii) what is the nature of morphological operations? (§2.1.3), and (iv) how direct is the relationship between meaning/function and form? (§2.1.4).

2.1.1 The level of the word: Lexicalism vs. non-lexicalism

We assume that a core computational system, the Syntax, builds (at least) linguistic expressions of phrasal/clausal size. As articulated in §1, there may also be a separate computational system, the Morphology, that is responsible for smaller, more-tightly knit linguistic expressions (words). The most basic dimension of variation across theories of the morphology/syntax interface is about the point of contact between the Syntax and the Morphology.

The classic picture of the morphology/syntax interface, as presented commonly in introductory linguistics courses, is that the Morphology builds words, and the Syntax takes these words and builds phrases/clauses out of them. There is something intuitive and easily graspable about this picture, and indeed this division of labor captures the empirical observation that phrasal syntax usually does not get to “see” inside words, in the sense of manipulating or selecting for subparts of words, etc. The idea that the Syntax and the Morphology interface at the level of the word is commonly referred to as lexicalism,⁵ and encapsulated in the LEXICALIST HYPOTHESIS. As articulated by

⁵For thorough discussion of different ways the term “lexicalist” has been conceived of over the years, as well as the related exoskeletal/projectionist vs. endoskeletal distinction, we refer the reader to Acedo-Matellán 2018 and Fábregas and Putnam 2020, Ch. 2.2.

Williams (2007, p. 353), the Lexicalist Hypothesis “suggests that the system of words in a language is independent of the system of phrases in a language in a particular way. It is independent of it, but communicates with it through a narrow channel—the ‘top-level’ properties of words. The system of words determines what the words of a language are and what their properties are. The system of phrases determines how words form phrases, based on the properties of words.” A family of frameworks have arisen around this hypothesis.

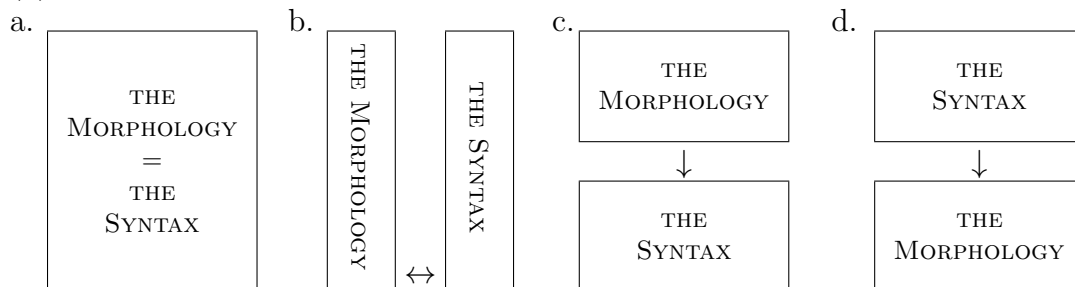
Alternatively, there are frameworks that *reject* the idea that morphology and syntax interface at the level of the word. Non-lexicalist frameworks are motivated by the intuition that there are too many deep similarities between linguistic expressions of word size and phrasal/clausal size, both within languages and cross-linguistically, for the Morphology and the Syntax to interface (only) at the level of the word. Non-lexicalism is a broader category than lexicalism, consistent with a number of positions about the point(s) of contact between morphology and syntax, e.g., morphology and syntax could interface at the level of something smaller than the word, they could interface at the phrasal level, or there could in fact be *no* distinction between them.

As we will see, the lexicalist vs. non-lexicalist distinction has broad repercussions within the landscape of theories.

2.1.2 The architecture of the grammar

Going beyond the basic lexicalist/non-lexicalist division (§2.1.1), we can ask about the *architecture* of the grammar—the timing relationship between the Morphology and the Syntax. Four basic possibilities have been explored prominently in the literature: the morphological computational system and the syntactic computational system could be one and the same, (4a), or they could be distinct, with the Morphology running *parallel* to the Syntax, *preceding* the Syntax, or *following* the Syntax, as illustrated respectively in the schemas in (4b-d), where arrows indicate the flow of information.

(4) Possible relationships between the Morphology and the Syntax



Of course, (4a-d) are snippets from larger architectures that also incorporate phonology and semantics, components of the grammar we sidestep in this Element, for the most part.

Let's first consider the inner two architectures, in (4b-c). The PARALLELIST schema in (4b) is intended to be understood in the following way: the Morphology and the Syntax operate wholly separately of each other, interfacing only at the *output* of both computations. Assuming the output of the Morphology is words, such parallelist approaches are inherently lexicalist.⁶ The schema in (4c) represents the classic PRE-SYNTACTIC picture of the morphology/syntax interface, where the output of the Morphology is the input to the Syntax. Again assuming the output of the Morphology is words, this architecture is also inherently lexicalist.

Now let's turn to the outer two architectures. The schema in (4a), which we will term a SYNTACTIC approach to the morphology/syntax interface, takes there to be only one computational system for linguistic expressions of all sizes. Syntactic approaches are therefore inherently non-lexicalist. Finally, in the schema in (4d), which we will refer to as a POST-SYNTACTIC approach to the morphology/syntax interface, the output of the Syntax is the input to the Morphology, which also is a non-lexicalist sort of interface. When discussing this last type of approach, we will sometimes refer to the Morphology as THE POST-SYNTAX, following the convention in the literature.

Within theories that posit some kind of morphological component, (4b-d), it is important to recognize that what one might consider to be a morphological phenomenon may or may not actually be dealt with by the morphological computational system. For example, in post-syntactic theories, (4d), the possibility emerges for words to be built in the Syntax, before the Morphology, with the Morphology handling other (aspects of) morphological phenomena. As another example, some lexicalist theories make a fundamental distinction between the status of inflectional and derivational morphology (see §1.1), with derivational morphology being the purview of the Morphology, and inflectional morphology being the purview of the Syntax; this is known as the Split Morphology Hypothesis (see discussion and references in Booij 1995a).

Off the bat, it may seem that a reduction of the Syntax and the Morphology to one computational system, as in (4a), is by far the most theoretically parsimonious option. But, as Chomsky put it in his early work on the morphology/syntax interface (see the beginning of §2.2), "it is to be expected that

⁶Other types of parallelist approaches are also possible, with more points of contact. Such approaches would *not* be lexicalist in the sense that we use the term here.

enrichment of one component of the grammar will permit simplification in other parts” (Chomsky, 1970, p. 13). The converse is true as well: if one component is simplified (or eliminated entirely), then it follows that another will need to be enriched. This push-and-pull is at the heart of morphosyntactic theorizing today.

2.1.3 The nature of morphological operations: Pieces vs. processes

The third fundamental point of variation across theories of the morphology/syntax interface—familiar to morphologists from Hockett 1954—revolves around the basic mode of generating (or otherwise accounting for the shape of) complex morphological forms, wherever this happens architecturally. One possibility is that morphological forms are related to each other through the addition of *PIECES* (discrete objects in the grammar). Another possibility is that morphological forms are related to each other by *PROCESSES* (“rules or formulas”, per Stump 2001, p. 1).⁷ These two possibilities boil down to a difference in what is stored, and how, in ways we will elucidate below.

But first, several terminological distinctions are in order. Morphologists generally distinguish *LEXICAL* elements from *FUNCTIONAL* elements, where *ROOTS* (e.g., *little*, *faucet*, *whirl*) are lexical and everything else (e.g., negation, case, tense, nominalization) is functional. As an independent property, a morphological element may be *BOUND*, unable appear on its own as a well-formed word, or *FREE*, not bound. (We will return to the free vs. bound distinction in §3.2.3.) Bound functional elements that precede or follow their stem are referred to as *AFFIXES*. Finally, we use the term *STEM* to pick out a morphological object that serves as the base for an affix or other bound functional morphology.

In a piece-based theory, all decomposable morphological elements—lexical and functional, free and bound—are stored independently of one another and listed with their own properties. The computational system has the job of combining these pieces, which end up *CONCATENATED*—placed one next to another—into a linear string, usually mediated by a hierarchical structure. For example, the complex word /kæbmz/ ‘cabins’, would be analyzed as the com-

⁷Stump (2001) refers to piece-based theories as “lexical”, and process-based theories as “inferential”. These terms sometimes get mixed up in the literature with other terms, especially with the confusing resemblance among “lexical” (= piece-based), “lexical” (= a root; not functional), and “lexicalist” (= a word-based interface between morphology and syntax). Hockett (1954) refers to piece-based theories as “item-and-arrangement”, and (some) process-based theories as “item-and-process”, also leading to potential confusion with respect to the meaning of the term “item”. As such, we do not use these more traditional terms here, opting instead for what we hope is more transparent terminology.

combination of the discrete, independently-stored pieces in (5a) and (5b), which come together to form (5c):⁸

(5) Piece-based approach

- a. /kæbm/ ‘cabin’
- b. /z/ ‘PLURAL’
- c. /kæbmz/ ‘cabins’

Most commonly, the pieces in a piece-based theory, like those in (5a,b), are called MORPHEMES, traditionally conceived as bundles of form and meaning/function that are not further decomposable, in the Bloomfieldian sense—though we will see throughout this Element that things are a fair bit more complicated than that. What really defines a piece-based theory is that complex morphological forms are the result of the combination of discrete and independently-stored concatenating pieces (at one or more abstract levels).

In a process-based theory on the other hand, one way of concretely modeling the computation is that it involves applying (morpho)phonological modifications (processes) to a stem. All bound functional morphology is accounted for in this way. Consider the rule for regular English plural marking in (6a) (where X represents a phonological string corresponding to a stem), and its application to the noun *cabin* in (6b).

(6) Process-based approach

- a. English plural rule: $X \rightarrow Xz$
- b. /kæbm/ ‘cabin’ $\rightarrow$ /kæbmz/ ‘cabins’

The rule in (6a) transforms the noun stem /kæbm/ into /kæbmz/. Notice that /z/ is not an independently stored element with the meaning ‘PLURAL’ (cf. (5b)), but rather is part of a phonological transformation associated with plurality. Process-based theories thus wholly eschew the traditional morpheme. In fact, affixation—a stem modification that involves adding segments before or after the stem, as in (6)—is just one type of possible modification that can be output by the rules in a process-based theory, alongside other modifications like ablaut and consonant mutation.

The piece-based vs. process-based distinction has implications for the relationship between morphological and syntactic computation (§2.1.1-2). In a process-based approach to morphology, the morphological and syntactic computations work fundamentally differently; this in turn implicates lexicalism

⁸Per convention, we usually represent functional elements in small caps, like PLURAL, or PL for short, and we enclose phonologically-represented forms in slashes, e.g. /x/.

and parallelist or pre-syntactic architectures. On the other hand, a piece-based approach to morphology could be pursued in a lexicalist or non-lexicalist approach, with any kind of architecture.

2.1.4 The (non-)independence of exponents: Realizational vs. incremental

A final crucial distinction among theories of the morphology/syntax interface is about the (in)directness of the relationship between meanings/functions and their related phonological forms, EXPONENTS. Traditionally, these are tied together, e.g., in a (piece-based) lexical entry like {PLURAL, /z/}. When they are unified in this way, information necessarily builds up incrementally as a complex word is built, with meanings/functions and phonological exponents in lockstep at all stages of the representation; this is why these theories are referred to as INCREMENTAL (Stump, 2001). Returning to our earlier example: there are two crucial stages in the formation of the complex word *cabins*, shown here as Stage X followed by Stage Y:

- (7) Incremental approach
 - a. Stage X: /kæbm/ ‘cabin’ (uninflected)
 - b. Stage Y: /kæbmz/ ‘cabins’ (= ‘cabin, PLURAL’)

There is a stage where the noun is uninflected, lacking any type of plurality, (7a), and one where it has a plural feature and a plural exponent, (7b). In other words, the root noun /kæbm/ ‘cabin’ becomes plural when /z/ ‘PLURAL’ is added. Importantly, there is no intermediate stage between these two stages of word formation, where some aspect of pluralness is present without the other.

On the other hand, many theories of the morphology/syntax interface divorce meanings/functions from their phonological exponent(s). In such theories, it is always the meaning/function (or some abstraction underlying the meaning/function) which in some metaphorical sense precedes or pre-exists the phonological exponents. These theories are referred to as REALIZATIONAL (Stump, 2001), because exponents *realize* meanings/functions that are already given.⁹ Returning again to the plural marking in *cabins*: the presence of a plural feature *first*, (8a), licenses the addition of /z/ *later*, (8b):

- (8) Realizational approach
 - a. Stage X: ‘cabin, PLURAL’
 - b. Stage Y: /kæbmz/ ‘cabins’ (= ‘cabin, PLURAL’)

⁹See also the Separation Hypothesis of Beard 1995.

In a fully realizational theory, there is a stage where the representation is wholly abstract (lacking exponents), and a later stage where the word is fully EXPONED, with phonological exponents matched to the abstract elements. As we discuss in §2.2-3, the most prominent theories of morphology today are realizational, whether lexicalist or non-lexicalist (§2.1.1).

Inherent to realizational theories is the need to *choose* the right phonological exponent in a given environment/context. Most realizational theories make crucial use of the ELSEWHERE CONDITION, also referred to as Panini’s Principle or the Subset Principle, which (in this context) compares possible phonological exponents and arbitrates in favor of (i) the most morphosyntactically-specific exponent that is (ii) compatible with the context (see discussion in Arregi and Nevins 2013). Put in another way, the best choice phonological exponent is the one associated with the *most* morphosyntactic information that also does not *conflict* with the morphosyntactic context. For example, consider the word *oxen* in English, which bears irregular plural marking: the regular plural exponent /z/ is not chosen (**oxes* in Mainstream American English) because there is a better match exponent, /ən/, that is associated with both PLURAL and the root *ox* (✓*oxen*).

With all of the concepts and dimensions introduced above in hand, we are now well-situated to dive more deeply into actual approaches to the morphology/syntax interface.¹⁰

2.2 Lexicalist approaches

In early generative grammar, the lexicon was taken to be a place for the storage of “formatives”, roughly equivalent to morphemes, where “each formative is [arbitrarily] assigned to various categories that determine its abstract underlying [phonological] form, the syntactic functions it can fulfill, and its semantic properties” (Chomsky and Halle, 1968, p. 7). Formatives constituted the input to the syntactic computational system, as the basic units over which the Syntax operated. Formatives like {‘cabin’, /kæbm/} and {PLURAL, /z/}—represented very informally here for ease—could thus be combined in the Syntax, together constituting an N (noun) node. This early approach was, for the most part, non-lexicalist, piece-based, and incremental.¹¹

¹⁰We refer the reader to Stewart 2016 for discussion of further parameters of variation across morphological theories.

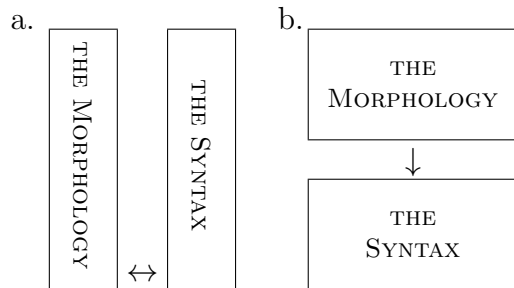
¹¹Chomsky 1957 is also non-lexicalist and (for the most part) piece-based, but realizational due to the nature of rewrite rules as the basic computational operation of the Syntax. This and other components of Chomsky’s early work were very similar to today’s prominent non-lexicalist approaches, which will be discussed in §2.3.

Chomsky’s (1970) *Remarks on Nominalization* subsequently argued that the pre-syntactic lexicon not only functions as a storage bin of basic, decomposed units, but also includes some complex morphological forms. Specifically, Chomsky proposed that idiosyncratic, “derived” nominals like *refusal* are inserted into the Syntax directly (i.e., must be stored or somehow generated in the lexicon), while transparently deverbal “gerundive” nominals like *refusing* are generated in the Syntax. Alongside similar sentiments developing in Halle 1973, this partial separation of complex morphological forms from the Syntax marks the birth of the Lexicalist Hypothesis (see §2.1.1), which more expansively encompassed the idea that there is a dedicated, non-syntactic system in the grammar that deals with the properties and behaviors of words.

A diverse array of approaches to the morphology/syntax interface adopt the Lexicalist Hypothesis in some form (see Scalise and Guevara 2005 and references therein). These approaches are naturally tuned into the differences between word-internal phenomena and word-external phenomena, hence motivating their separation in the grammar. A corollary of the Lexicalist Hypothesis is the Lexical Integrity Hypothesis, which holds that the Syntax cannot manipulate sub-word pieces (see Lieber and Scalise 2006 for an overview).

Lexicalist approaches to the morphology/syntax interface generally make use of one of the two (sub-)architectures schematized in (9) (excerpted from (4) in §2.1.2):

(9) Architectures commonly found in lexicalist approaches

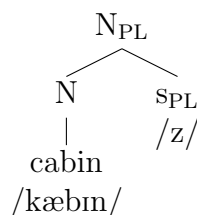


The first is a parallelist architecture, and the second is a pre-syntactic architecture. With respect to the Y model in §1.2, it is point of contact 1 that is most relevant for morphology in pre-syntactic approaches, while parallelist approaches account for morphology outside of the Y model.

Some lexicalist approaches are piece-based, with complex words resulting from the combination of discrete elements (§2.1.3). For example, Di Sciullo and Williams (1987) propose that the morphological component is wholly separate from the syntactic component, but operates largely the same, in terms of

(i) being based around pieces: morphemes for the Morphology, words for the Syntax, and (ii) building hierarchical structures, via affixation and compounding in the Morphology, phrase-building in the Syntax. The Morphology and the Syntax run separately but in parallel, as in the schema in (9a), with the Syntax operating over purely abstract atoms. Actual words that have been generated by the Morphology are inserted “late” into the Syntax, into head positions. Returning to /kæbmz/ ‘cabins’: this complex word would be generated in the Morphology by structurally combining {‘cabin’, /kæbm/} and {PLURAL, /z/}.

(10) *cabins* in Di Sciullo and Williams 1987



This complex noun can then be inserted into an appropriate N node in a syntactic representation. This approach is incremental within the Morphology, but realizational in terms of the relationship of fully-formed words to the output of the Syntax. Other piece-based lexicalist theories include the incremental theories Lexical Phonology and Morphology (Kiparsky, 1982) and Minimalist Morphology (Wunderlich, 1997), and the realizational theory Harmonic Serialism Morphology (Müller, 2021); all these theories have a pre-syntactic architecture like that in (9b).

While there are piece-based lexicalist approaches, the most prominent lexicalist theories today are process-based, with complex morphological forms related to each other by rules (again see §2.1.3). Indeed, if the Morphology is independent from the Syntax, this opens the door for morphology to be radically different from syntax in this way, with the Morphology operating via transformations, rather than through the combination of pieces. Accordingly, in process-based approaches, morphemes and morphological structure are both rejected, and hence these approaches are “a-morphous” (Anderson, 1992).

Most process-based theories afford a special primitive status to the notions of LEXEME and PARADIGM. A lexeme is a basic stored unit representing one lexical meaning and syntactic category, like CABIN (noun).¹² All lexemes have a paradigm, defined as the set of all possible combinations of contrastive inflectional features that apply to a lexeme (based on its syntactic category),

¹²We represent lexemes in all caps.

	PRESENT		PAST		PARTICIPLE	
	SG	PL	SG	PL	SG	PL
1	/it/	/it/	/ert/	/ert/	/itən/	/itən/
2	/it/	/it/	/ert/	/ert/	/itən/	/itən/
3	/itz/	/it/	/ert/	/ert/	/itən/	/itən/

Table 1: Paradigm for the lexeme EAT (verb)

as well as the corresponding actual word forms that fill out that paradigm. Consider verbs in English, with contrastive features including tense/finiteness (PRESENT, PAST, and PARTICIPLE, at least) and subject agreement (1, 2, 3 for person and SG, PL for number). The lexeme EAT (verb) has the paradigm in Table 1 in Mainstream American English, with words given in their underlying phonological forms. The 18 cells defined by the contrastive inflectional features for English verbs are filled, in the case of the lexeme EAT (verb), by 4 distinct word forms, *eat*, *eats*, *ate*, *eaten* (/it/, /itz/, /ert/, /itən/).

Paradigms are highly language-specific, and different lexemes within a language can fill out a paradigm in different ways. Most verb lexemes in English have fewer distinct forms than EAT (verb) does, e.g., DANCE (verb) only has three: *dance*, *dances*, *danced*, with no distinction between the PAST form and PARTICIPLE form. But, the copula BE has more distinct forms: *am*, *is*, *are*, *was*, *were*, *been*, with further person/number distinctions. A language may alternatively have no inflectional morphology and thus paradigms with only one cell, or they may have many inflectional distinctions and thus extremely large and complex paradigms.

WORD-AND-PARADIGM approaches to morphology take words and the paradigms that they occupy to be crucial theoretical objects. The best-known Word-and-Paradigm approach today is PFM (Stump, 2001, 2016), though this has many important intellectual precursors (including Matthews 1972; Zwicky 1985; Anderson 1992; Aronoff 1994). In PFM, a cell in a paradigm represents a complete morphosyntactic property set, namely the set of features that uniquely pick out that cell. In Table 1, for example, the top rightmost paradigm cell corresponds to the morphosyntactic property set {1, PL, PARTICIPLE}. The morphosyntactic property set in a cell licenses morphophonological transformations on the “root” (phonological stem) of a lexeme, thereby accounting for the actual word form in that cell of the paradigm. Word-and-Paradigm approaches are thus realizational in nature—the features and lexeme are separate from, and in an important sense prior to, the phonological expo-

nents of them.¹³ PFM has a parallelist architecture, with word forms mapped to positions in the syntactic structure *after* the Syntax (Stewart and Stump, 2007).

To give a more complete sense of how PFM works, let’s work through one (somewhat simplified) example. Formally, a PARADIGM FUNCTION is a mapping from a lexeme, like EAT (verb), to the word form in a particular cell of the lexeme’s paradigm, like /itən/ in the cell {1, PL, PARTICIPLE}. Paradigm functions break down into one or more block(s) of REALIZATION RULES (RRs) that act on the phonological stem of a lexeme and give rise to (potential) morphophonological changes, in the context of a particular morphosyntactic property set (σ). For English verbs, there is only one morphological “slot” for inflectional material, so the paradigm function consists of just one block of RRs. The Elsewhere Condition (see §2.1.4) determines which RR applies, given the lexeme and morphosyntactic property set. The relevant RR for the case at hand—*eaten*—is given in (11).

$$(11) \quad \text{RR}_{\{\text{TNS:PART}\},\text{V}}(<\text{X},\sigma>) = <\text{X}\text{ən}',\sigma>$$

The RR in (11) states that it is licensed in the context of any morphosyntactic property set containing participial features (TNS:PART), for the category verb (v). The RR further states that, when applied to the phonological stem of a lexeme, X, with morphosyntactic property set σ , the form $\text{X}\text{ən}'$ is licensed. For EAT (verb), the phonological stem would be /it/, and (11) licenses the transformation of /it/ to /itən/ in the context of (for example) the morphosyntactic property set {1, PL, PARTICIPLE}. This RR would of course need to be restricted to certain lexemes, as the *-en* participle-forming strategy is not fully productive in English. Other competing RRs within the same rule block would license other verb forms in a similar way.¹⁴

Beyond the theoretical parameters discussed above, there are myriad other ways that lexicalist theories differ from each other, and myriad other lexicalist theories. WEAK lexicalist theories separate inflectional morphology from derivational morphology, with the latter being the purview of the Morphology and the former being the purview of the Syntax or post-syntax. STRONG lexicalist theories locate both inflectional and derivational morphology in the morphological component of the grammar, but attribute distinct roles to them:

¹³While most process-based theories are realizational (like PFM), such theories may be incremental as well, as in Articulated Morphology (Steele, 1995).

¹⁴For a form like /ert/ (EAT (verb), PAST), where it looks like it is the first vowel of EAT’s phonological stem that has been altered, there are a few possible analytical routes afforded by PFM, including an ablaut-inducing RR, or the selection of a distinct phonological stem in this context.

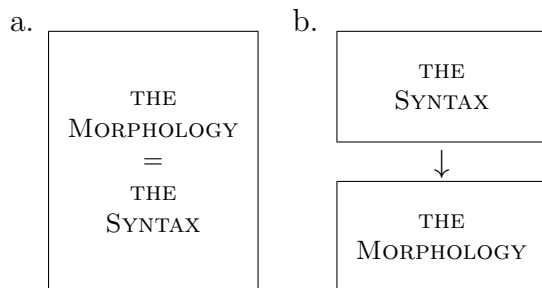
derivational morphology relates different lexemes to one another, while inflectional morphology relates different forms of a particular lexeme to the base of the lexeme; this is the case, for example, in PFM. (See Booij 1995a and references therein on the Split Morphology Hypothesis.)

Lexicalist theories also differ in the way the morphological component is structured. A number of lexicalist theories organize the Morphology into LEVELS or STRATA, with different sorts of morphology (and phonology) applying over different—and crucially ordered—subsets of a language’s morphological system (Siegel 1974; Pesetsky 1979; Kiparsky 1982, a.o.). Yet other lexicalist theories model the morphological component as a hierarchical network. For example, in Network Morphology and Information-based Morphology, morphological information is organized around major morphosyntactic categories, with nodes in the network encoding shared properties across a morphosyntactic category and (deeper in the network) subgroups of that category, all the way down to individual lexemes (terminal nodes in the network); see, e.g., Corbett and Fraser 1993; Brown and Hippisley 2012; Crysmann and Bonami 2016. A hierarchical network also plays a central role in Construction Morphology (Booij 2010) and related frameworks (e.g., Jackendoff and Audring 2020). Construction-based approaches to grammar expand such a network outwards to include even syntax, a sort of *hyper*-lexicalism, which is a radical alternative to the non-lexicalist approaches we turn to next.

2.3 Non-lexicalist approaches

Non-lexicalist approaches are generally motivated by the similarities between what happens within words and what happens in larger linguistic expressions, thereby taking morphological and syntactic computation to operate with the same kinds of principles and processes, at least to a significant degree. The relevant schematic architectures are excerpted from (4) in §2.1.2 and repeated below.

(12) Architectures commonly found in non-lexicalist approaches



The privileged role of the Syntax in both architectures in (12) determines some major characteristics of all non-lexicalist approaches. Since linguistic expressions of all sizes originate within the syntactic computational system, the building blocks of words must be fundamentally the same as the building blocks of larger linguistic expressions. And, since all (generative) theories of syntax today are piece-based, with the computation involving the combination of discrete objects into hierarchical structures, it follows that non-lexicalist morphological theories are also piece-based and hierarchical. What are the pieces that the Syntax operates over in a non-lexicalist framework? For some theories, this is the role of the (abstract) morpheme. Other theories posit a yet smaller, more abstract unit, the feature.

The most fundamental distinction among types of non-lexicalist theories is, of course, that depicted in (12) above: whether the Syntax is the *only* computational system, or whether there is at least a limited separate morphological computational system after the Syntax, often referred to as the post-syntax. The former approach denies the exceptionality of morphology entirely and is (by necessity) willing to complexify the Syntax to accommodate morphological phenomena. The latter approach aims to capture both the fundamental similarities between syntax and morphology as well as their differences, thereby maintaining a simpler syntactic computational system.

In practice—and as is obscured by (12)—syntactic and post-syntactic approaches actually lie on a continuum, (13):

(13) Continuum of syntactic and post-syntactic approaches



Moving from the left to the right on the continuum is a move towards more post-syntactic computation. Where a theory falls on the continuum determines whether it is realizational or incremental (see §2.1.4 on this distinction), as demarcated by the point labeled “late insertion”. Approaches to the left of this point forgo all non-syntactic computation, and therefore phonological forms must be present in the Syntax, as part of the building blocks of the Syntax; these approaches are thus incremental and based around traditional morphemes (pairings of form and meaning),¹⁵ with morphology accounted for at point 2 of the Y model in §1.2.

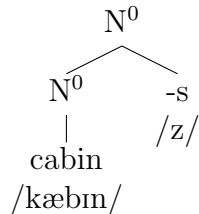
The most minimal deviation from a syntax-only account is to introduce

¹⁵An exception, perhaps, is Bruening’s (2017) Consolidated Morphology, a purely syntactic approach to morphology that incorporates some aspects of realization into the Syntax.

phonological exponents *after* the Syntax, i.e., to be realizational instead of incremental. In this context, being realizational means the Syntax operates over purely abstract elements, lacking phonological forms. LATE INSERTION refers to the computation that involves choosing and inserting phonological exponents, given the output of the Syntax. The most minimal post-syntax, then, is *just* late insertion, with all approaches to the right of this point on the continuum having late insertion as well as further (types of) post-syntactic morphological computation.

Let’s turn first to some of the syntax-only, incremental, morpheme-based theories that occupy the far left of the continuum in (13). One of the earliest approaches that moves away from lexicalism, by Lieber (1988, 1992), aims to develop “one general set of principles [...] that allows the generation of both well-formed words and well-formed sentences” (Lieber, 1992, p. 21). Lieber adopts X-bar syntax (Jackendoff, 1977; Chomsky, 1981), with some modifications, in particular, that the X^0 (head) level can recurse, allowing for X^0 to combine with another head or phrase and project another X^0 . Under this proposal, our usual example, *cabins*, is a complex N^0 containing the root *cabin* as the minimal head, combining with the plural suffix *-s* (which lacks its own categorial features) and projecting a larger N^0 head. Thus, Lieber takes what we call a NON-INTEGRATED approach to word-internal morphemes—they are merged directly inside a complex X^0 :

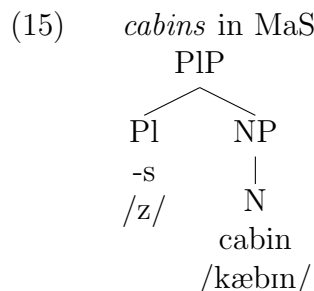
(14) *cabins* in Lieber 1992



The highest X^0 (not dominated by any further heads) delineates a word from the rest of the syntactic structure, and these highest X^0 s (words) for the most part act as atomic units for the rest of syntactic structure-building. Kiparsky (2021) takes a similar non-integrated approach, with “morphology as the innermost layer of syntax” (p. 393). This is a lot like how words work in piece-based lexicalist approaches—compare (14), for example, with Di Sciullo and Williams’ structure in (10).

Other purely syntactic theories take an INTEGRATED approach: all morphemes, even bound/affixal ones, are run-of-the-mill heads of phrases, incorporated into the functional spine. Collins and Kayne’s (2023) recent *Morphology as Syntax* (MaS) exemplifies an integrated, purely syntactic approach to mor-

phology. They would posit something like (15) for *cabins*:



The root, *cabin*, is the head of NP, and the bound plural morpheme *-s* is the head of its own phrase, PIP. Given that the two morphemes in *cabins* are heads of separate phrases, their status as forming a word must either be (i) epiphenomenal, or (ii) derived by something bringing the two morphemes together into a tighter unit; see §3.3. Further, MaS is a CARTOGRAPHIC approach, holding that all languages share a universal set of features, and these features get mapped onto the syntactic structure in a universal sequence (Rizzi, 1997; Cinque, 1999), each feature as its own head. Relatedly to the cartographic explosion of featurally-simplex heads, it is posited in MaS that heads can be null under specific morphosyntactic licensing conditions. As we will discuss at many points in §4, MaS’s cartographic, incremental approach, eschewing all post-syntactic mechanisms, significantly constrains possible analyses of empirical phenomena.

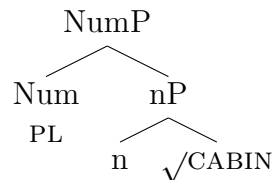
Although there are some purely syntactic approaches to morphology, like those described above, today nearly all non-lexicalist theories fall somewhere further to the right on the continuum in (13), i.e., they involve some kind of post-syntactic computation. There is a two-fold motivation for not pursuing a purely syntactic approach: (i) to simplify the Syntax in terms of its computational possibilities, and (ii) to reap the empirical benefits of choosing exponents “late”, i.e., of having a realizational theory (to be discussed extensively in §4). Recall from §2.2 that the overwhelming majority of lexicalist theories, too, are realizational, putting otherwise quite different theories in sync with each other on this question. In terms of the integratedness of morphology into syntax (see definition above), post-syntactic approaches are generally integrated, as this allows for a more direct connection between morpheme order and syntactic processes, including capturing Mirror Principle effects (§1.1).

The non-lexicalist approach that dominates the theoretical landscape today is DM (Halle and Marantz 1993, 1994; see also references in fn. 4). This label is an umbrella over a set of approaches that share two definitional tenets: (i) syn-

tax first,¹⁶ and (ii) late insertion. The first tenet—syntax first—is a statement about the architecture: the Syntax is the basis for morphology, or put another way, morphological phenomena start with/in syntax. The second tenet of DM—late insertion—refers to the realizational nature of the framework, with the Syntax operating over purely abstract elements, which in DM are abstract morphemes (e.g., a feature, set of features, or root). Phonological exponents are introduced through an operation referred to as VOCABULARY INSERTION, which maps abstract morphemes—the heads in the syntactic structure—to phonological exponents, subject to the Subset Principle (see §2.1.4). This framework gets its name from the idea that aspects of morphology that lexicalist approaches attribute to one module (the Morphology, see §2.2) are instead accounted for *across* the grammar, in fact, occurring at all labeled points in the Y model in §1.2.¹⁷

To better understand the basics of how DM works, let’s run through the usual example, *cabins*. In line with the syntax-first tenet of DM, a syntactic structure like that in (16) is generated, with abstract morphemes integrated fully into phrasal syntax. The structure in (16) incorporates the assumption that roots don’t have lexical categories, and so they necessarily combine with a categorizing head; in (16), nominalizing “little n” selects the root and projects an nP. The Num(ber) head, which contains the feature PL, selects nP and projects a NumP.

(16) Syntactic structure for *cabins* in DM



The syntactic structure in (16) is passed to the morphological component, which (minimally) consists of Vocabulary Insertion. The relevant vocabulary items that are accessed are given in (17), each a correspondence between a particular abstract element and a particular phonological exponent:

¹⁶This tenet is often called “syntax all the way down”, referencing the fact that the Syntax provides the basis for morphological phenomena, i.e., all morphology starts with syntax. But, as this phrasing could be misconstrued to mean that the Syntax explains *all* of morphology (or that all of morphology is syntax), we opt for the hopefully clearer “syntax first” characterization for post-syntactic approaches.

¹⁷While DM is adopted almost uniformly in the context of Minimalist syntax, a recent variant of DM has been developed in the context of LFG, called Lexical-Realizational Functional Grammar (Melchin et al. 2020; Asudeh et al. 2024; et seq).

(17) Vocabulary items for *cabins* in DM

- a. $\sqrt{\text{CABIN}} \leftrightarrow /kæbm/^{18}$
- b. $n \leftrightarrow /∅/$
- c. $PL \leftrightarrow /z/$

The vocabulary item that is the best match for a given node in the structure is chosen according to the Subset Principle. Assuming that both *n* and *Num* somehow form a word with the root and are linearized after the root (see §3.3), the form derived by the morphological component is */kæbm-∅-z/*.

DM approaches differ from each other in terms of the richness of the post-syntax. In some varieties of DM, Vocabulary Insertion is essentially the *only* computation that happens in the post-syntactic morphological component (e.g., Trommer 1999). But, most varieties of DM have a richer post-syntax. Some include just a few operations that can alter the output of the Syntax (e.g., Embick and Noyer 2001; Harizanov and Griбанова 2019; Myler 2024a). For others, the morphological component has a rich structure and a large set of operations (e.g., Halle and Marantz 1993; Embick 2010; Arregi and Nevins 2012). Some commonly proposed post-syntactic mechanisms, i.e., mechanisms in the Morphology, include: Lowering, which adjoins a syntactic head to the head of its complement; Local Dislocation, which trades a morphological adjacency relationship for an attachment relationship (potentially resulting in reordering); Impoverishment, which deletes one or more features from the morphosyntactic representation; Fusion, which combines two morphemes into one; Fission, which splits one morpheme into two; and Dissociated Node Insertion, which inserts a (new) morpheme. All of these mechanisms will be discussed at one or more points in §3 and §4 of this Element.

An alternative post-syntactic approach that has gained prominence in recent years is Nanosyntax (see Caha 2009; Starke 2009, 2011). Nanosyntax shares some basic properties with DM—syntax first and late insertion—but involves a radical re-thinking of the interplay between syntax and morphology. Like MaS, Nanosyntax is cartographic, assuming terminal nodes in the Syntax at the level of the feature. In addition, Nanosyntax has *no* post-syntactic mechanisms apart from late insertion.

Putting the two basic Nanosyntactic assumptions together—heads as fine-grained atomic features, and late insertion as the *only* post-syntactic mechanism—leads to the proposal that insertion of exponents can target something larger

¹⁸Early in DM theorizing, root morphemes were assumed to have phonological forms in the pre-syntactic lexicon, i.e., roots were not subject to late insertion. However, it is now generally assumed in DM that all morphemes are abstract and subject to late insertion. See Harley 2014 and responses to this work, as well as Bobaljik 2012.

	Realizational		Incremental	
	Piece-based	Process-based	Piece-based	Process-based
Lexicalist	Harmonic Serialism Morphology	PFM	Minimalist Morphology	Articulated Morphology
Non-lexicalist	DM, Nanosyntax	(unattested)	Morphology as Syntax	(unattested)

Table 2: Typology of theories

than the syntactic head. In Nanosyntax, it is syntactic constituents (phrases) that are mapped to exponents, via a mechanism referred to as PHRASAL SPELL-OUT. Phrasal spellout, in turn, necessitates a number of other modifications to morphological/syntactic theory, including (i) the use of the Superset Principle (in lieu of the otherwise-ubiquitous Subset Principle, §2.1.4), which states that a given vocabulary item can be inserted just in case it *contains* the structure it is spelling out, (ii) an Elsewhere Condition that prefers the *smallest* matching structure that satisfies the Superset Principle, and (iii) an algorithm that allows the Syntax and phrasal spellout to “loop”, with the available vocabulary items in a language (called “morphemes” in Nanosyntax) thus able to influence the syntactic derivation. For overviews of Nanosyntax, we refer the reader to Baunaz and Lander 2018 and Caha et al. 2024.

There are a number of other non-lexicalist approaches to the morphology/syntax interface. On the more syntax-y side, these proposals include Spanning (Svenonius, 2012, 2016; Merchant, 2015), Mirror Theory (Brody, 2000), and Borer’s Exoskeletal Model (Borer, 2005a,b, 2013). On the more phonology-y side, marrying aspects of DM with Optimality Theory, are Distributed Optimality (Trommer, 2001) and OT-DM (Rolle, 2020). Eschewing OT, some approaches propose a simple “translational” relationship between syntax and phonology (Scheer 2016, Newell 2021), necessarily making use of enriched and highly abstract phonological representations.

2.4 Section summary

This section covered a lot of theoretical ground, introducing key theoretical concepts at the morphology/syntax interface that will enable readers to approach a diverse array of literature. Table 2 above takes the most important theoretical dimensions and uses them to define a theory space, populating this “paradigm” of approaches with some of the theories discussed in this section, with gaps where types of theories are unattested, specifically in the absence of non-lexicalist process-based theories. In the coming sections, we show how these theories and theoretical dimensions map to the empirical landscape.

3 On words and word formation

Given the centrality of the notion “word”, both as a defining feature of lexicalism and as delineating the empirical domains of morphology and syntax, in this section we consider the status of words as constituting a particular type of linguistic expression. We begin in §3.1 by surveying the range of word size/complexity we find within and across languages. In §3.2, we discuss the widely-utilized distinction between grammatical/morphosyntactic words and phonological/prosodic words, as well as issues in defining wordhood. In §3.3, we turn to the theoretical mechanisms (or lack thereof) that have been proposed to generate morphosyntactic words in different types of approaches.

3.1 What do words look like? Small words and big words

This section examines words of different sizes and draws out related empirical generalizations and theoretical implications. §3.1.1 focuses on variation in morphological complexity. §3.1.2-4 explore strategies for forming large words—bound functional morphology, compounding, and noun incorporation—leading into a more in-depth discussion of polysynthesis in §3.1.5. §3.1.6 considers how different types of theories approach this empirical diversity.

3.1.1 Word complexity across languages

A common way to define morphological complexity is the ratio of morphemes (or exponents) to words in a linguistic expression. Morpheme-to-word ratios are useful in taking a birds-eye view of morphological patterns within a language, as well as variation cross-linguistically. Consider the English example in (18), a rather typical sort of English sentence:

(18) The dish-wash-er has clear-ly be-en broke-n for year-s.

In (18) there is both a relatively low ratio of morphemes to words (14:8, so averaging 1-2 morphemes per word) and no individual word is very complex.

Morpheme-to-word ratios form a continuum across languages. On one end of the continuum, languages have morpheme-to-word ratios that are close to 1, like Yorùbá (Niger-Congo):

(19) Èmi ni mo máan kọ ọ ní isé
1SG FOC 1SG IPFV teach 2SG THM work
‘It is I who used to teach you to work.’ (Yorùbá, Adebayo 2020, 21)

Languages that generally have single-morpheme words, like Yorùbá, are referred to as ISOLATING languages in the typological literature. Nevertheless, there usually (or perhaps always) are some processes that do form multi-morphemic words even in isolating languages, and this is indeed true for Yorùbá (Adewole, 1995); see §3.1.3. At the other far end of the continuum are languages with *very* high morpheme-to-word ratios, often called POLYSYNTHETIC languages, which will be discussed more in §3.1.5.

In between these extremes are SYNTHETIC languages, such as Lithuanian (see (2)), where complex words are the norm, but unlike in polysynthetic languages, these words do not usually stand in for whole sentences. Languages termed synthetic generally have robust inflectional marking, e.g., case, tense, and agreement. English, then, turns out to be somewhere in between synthetic and isolating, as even though English has robust derivational morphology, it has very little inflectional morphology. Such languages are sometimes termed ANALYTIC in the typological literature.¹⁹

3.1.2 Bound functional morphology

One way to build complex words is through bound functional morphology. Most commonly, this is found in the form of AFFIXATION, which canonically refers to segments before or after a stem that signal some kind of morphological contrast. A PREFIX is an affix that precedes its stem; a SUFFIX is an affix that follows its stem. The large morpheme-to-word ratio in the example from Ch’ol (Mayan) in (20) is achieved entirely via affixation, with two complex words built around the roots *choñ* ‘sell’ and *wäy* ‘sleep’ (both bolded).

- (20) Ta’ k-**choñ**-b-e-yety-la k-**wäy**-ib-tyak.
 PFV 1ERG-sell-APPL-DTR-2ABS-2PL 1POSS-sleep-NMLZ-PL.INDF
 ‘I sold you (pl.) some of my beds.’ (Ch’ol, Shenia Arcos Álvaro, p.c.)

In (20), the prefix *k-* indicates that the subject of the verb and the possessor of the object are each 1st person. A series of suffixes follow both roots. The suffixes closest to the root ‘sell’, namely *-b* and *-e*, are valency changing, i.e. they modify the number of arguments the verb can take, in this case allowing the transitive root *choñ* to become ditransitive and specify a recipient. The suffixes *-yety* and *-la* indicate that the recipient, which has absolutive case, is 2nd person and plural. The root ‘sleep’ also bears suffixes: *-ib* is a nominalizer,

¹⁹Note that the term “analytic” is not used consistently across the literature: sometimes it is used interchangeably with “isolating”, sometimes it picks out its own stretch of the continuum, and sometimes it subsumes both isolating and English-type languages.

deriving the noun ‘bed’, and *-tyak* marks this noun as an indefinite plural.

There are many other types of bound functional morphology that go beyond just straightforward prefixation and suffixation, such as infixation (the appearance of morphology intruding inside a stem) and circumfixation (morphology that surrounds a stem), as well as so-called morphological processes, like ablaut, reduplication, and consonant mutation (see, e.g., Inkelas 2014, ch. 3). Process-based theories (see §2.1.3, §2.2) incorporate morphological processes as just another kind of transformation of a stem, on par with the transformations that result in prefixation or suffixation. Piece-based theories on the other hand (see §2.1.3 et seq) are committed to analyzing *all* types of bound functional morphology as (essentially) just prefixes or suffixes. For example, Kalin (2022) argues that infixes start out as prefixes or suffixes and then displace into their stem. Along similar lines, it has been argued that circumfixes always decompose into a separable prefix and suffix (see Marušič 2023 and references therein). In a piece-based theory, apparent process morphology necessitates underspecified and highly abstract phonological representations (e.g., Lieber 1992) and/or the interaction of affixes with other (morpho)phonological processes (e.g., Bye and Svenonius 2012).

Greenberg (1966), in his foundational typological work, observed that derivational morphology typically appears closer to the root than inflectional morphology (Universal 28). The markers in (20) that pertain to person, number, case, and definiteness are clear cases of inflectional affixes. The clearest example of a derivational affix is the nominalizer, because it turns a verb root into a noun. The affixes that pertain to argument structure and valency are harder to categorize. In the complex noun in (20), derivational *-ib* appears closer to the root than inflectional *-tyak*. And, in the complex verb in (20), the valency-changing suffixes are closer to the root than the purely inflectional person/number suffixes.

The observation that inflectional morphology is more peripheral than derivational morphology has played a central role in certain morphological theories. Lexicalist word-and-paradigm theories (both weak and strong; see end of §2.2) predict derivational morphology should always be closer to the root, because of its distinct role/location in the grammar compared to inflectional morphology. In non-lexicalist theories, the inflectional/derivational distinction carries less theoretical weight and is often used only as a descriptive shorthand. This is because, in non-lexicalist theories, both types of morphology correspond to syntactic heads.²⁰ Indeed, the inflectional/derivational distinction in this

²⁰Note, however, that some work in DM has proposed that certain types of morphemes are introduced post-syntactically, e.g., theme vowels and agreement (Halle and Marantz, 1993;

type of theory may be entirely epiphenomenal, deriving (as a tendency) from structural closeness to the root, with closeness resulting in more idiosyncrasy, interaction with root meaning, etc.

In this context, it is important to note that there are plenty of known exceptions to Greenberg’s generalization. There are cases where inflectional morphology is closer to the root than derivational morphology (see discussion in Booij 2006, p. 659-660). A single type of functional morphology might also have divergent behaviors, sometimes seeming inflectional and other times derivational (e.g., see Harley 2008a and references therein on the causative in Japanese). Along similar lines, some argue that there is an inflection-derivation continuum as opposed to a strict divide (Booij 1995a; Haspelmath 1996; a.o.).

3.1.3 Compounding

COMPOUNDING describes the process of combining two roots (or stems) into a word. Compounding is often regarded as the most widespread morphological phenomenon, partly due to its presence even in typically isolating languages. In Yorùbá (cf. (19)), for example, the noun meaning ‘school’ is a compound, *ilé-ìwé*, from ‘house’ and ‘book’ (Adebayo, 2020, p. 20).

Cross-linguistically, noun-noun compounds are particularly common (Guevara and Scalise, 2009). The Dutch (Germanic) example in (21) features a very large compound entirely made up of nouns, displaying the characteristic recursion found in compounding: elements in compounds can themselves be compounds.

- (21) een [[wet-geving]-s-[advies-commissie]]-[voortgang-s-rapportage]
 DET law-giving-LNK-advice-committee-progress-LNK-report
 ‘a legislative advisory committee progress report’ (Dutch)

As indicated by the bracketing in (21), the interpretation involves crucially nested structures, containing compounds meaning ‘legislative’, ‘advisory committee’, ‘legislative advisory committee’, and ‘progress report’. This example also illustrates a typologically common phenomenon whereby the components of a compound are connected by an apparently semantically vacuous linking element, which is *-s* in this example (see also §4.5.2).

Compounds have a number of specific (sets of) properties cross-linguistically. Compound forms may be comprised of two or more roots/stems from a variety of lexical categories, but they generally convey a single concept. Inflection usu-

Embick and Noyer, 2007). Such morphemes are termed ORNAMENTAL or DISSOCIATED. See also §4.5.2.

ally does not appear inside of compounds (though this is not exceptionless), but rather only on the compound as a whole. The meaning of a compound is typically at least weakly compositional (Pirrelli, 2002), with the meaning of all the components contributing to the meaning of the whole, though with the mediating relationship(s) among the components variable across compounds, e.g., *nighttime* is a time *at* night, while *tequila time* is a time *for* tequila. But, sometimes the meaning of a compound is quite opaque and idiomatic, e.g., as in the English compounds *cocktail* and *hogwash*. Lexicalist approaches take (the possibility of) idiomaticity in compounds as evidence of their being formed outside of the syntactic component of the grammar, and thus corresponding to a listed meaning. Others counter with the observation that idioms/listedness exist at the phrasal/syntactic level as well (e.g., Di Sciullo and Williams 1987; Marantz 1997), and so this cannot be taken as a criterion for diagnosing the exceptionality of the word level. Nevertheless, most morphological theories agree that compounds have hierarchical structures, even theories that do not attribute structure to non-compounding word-formation processes (e.g., as in Anderson 1992).

See Harley 2009, Bauer 2017, Punske 2016, and citations therein for further discussions of different types of compounding, as well as myriad other typological and theoretical questions that arise in the area of compounding.

3.1.4 Noun incorporation

NOUN INCORPORATION, like compounding, involves the combination of independent roots or stems into a single complex unit, but specifically, a predicate incorporating one of its arguments. The example in (22) compares a Nahuatl (Uto-Aztecan) clause for which the object is a free-standing noun phrase to one in which the notional object is incorporated into the verb stem to form the predicate.

- (22) a. ni-k-na:maca **tlaxcal**-li
 1SG.SBJ-3SG.OBJ-sell tortilla-ABS
 ‘I sell tortillas.’
 b. ni-**tlaxcal**-na:maca
 1SG.SBJ-tortilla-sell
 ‘I sell tortillas.’ (Lit. ‘I tortilla-sell’)
 (Nahuatl, Launey 2011, p. 170)

Launey (2011) notes that (22a) is more appropriate as an answer to the question “What are you selling?” whereas the example with noun incorporation (22b) would be more felicitous as a response to a question such as “What do

you do?” Cross-linguistically, noun incorporation is often associated with stative predicates and lends itself to habitual or generic interpretations (Chung and Ladusaw, 2004; Farkas and de Swart, 2003; Dayal, 2011). And, whereas compounds can have opaque or idiomatic meanings, noun incorporation usually yields a more transparent interpretation, though it may still be only weakly compositional (Mithun 1984; McKenzie 2022).

Incorporation of a nominal also has syntactic consequences for the clause. For example, the verb in (22b) is only notionally transitive, as it lacks the object marker *k-* that is present in (22a); in other words, incorporation has resulted in the formation of an intransitive clause. Further, an argument incorporated into the predicate generally appears without any inflectional morphology (similar to the usual absence of inflection inside compounds); note, for example, that the incorporated object in (22b) lacks the absolutive marker *-li*, present in (22a).

There are robust cross-linguistic generalizations about what sorts of arguments can incorporate into a predicate. While noun incorporation most often describes an object incorporating into a verb, oblique arguments can be incorporated in many languages. In addition, a handful of languages with noun incorporation can also incorporate subjects. Importantly, however, subject incorporation is typically restricted to the sole internal argument of an intransitive predicate, i.e., incorporated subjects cannot be agentive (see e.g. Mithun 1984; Baker 1988; Polinsky 1990; and discussion in Massam 2009).

The restrictions on which arguments can incorporate, and especially the exclusion of transitive and agentive subjects, makes noun incorporation look like a syntactic effect: external arguments have too distant of a syntactic relationship to the predicate to incorporate into it. As a result, both lexicalist and non-lexicalist theories often take noun incorporation to be a syntactic process. The most well-known account of syntactic noun incorporation is probably that of Baker (1988), who proposes to analyze it as involving head movement of N into V. For non-lexicalist approaches, such word-formation in the syntax is expected. But, for lexicalist approaches, syntactic noun incorporation would constitute an exceptional type of case where the word level must make reference to syntactic structure. Thus, some lexicalist works argue that—despite its syntactic restrictions—noun incorporation is still a morphological process, at least in certain languages (e.g., Mithun 1984; Anderson 1992).

In line with syntactic approaches to noun incorporation, some languages exhibit a clearly phrasal form of this phenomenon, known as PSEUDO-NOUN INCORPORATION (following Massam 2001), e.g., Niuean (Polynesian), Ch’ol, Tzotzil (Mayan), and Hindi (Indo-Aryan). Many of the restrictions and interpretative effects observed in pseudo-noun incorporation closely resemble

those found in true noun incorporation. Accounts of pseudo-noun incorporation often rely on syntactic mechanisms (Massam, 2001; Coon, 2010) or post-syntactic mechanisms (Clemens, 2014; Clemens and Coon, 2018).

For much more discussion of the empirical profile and theoretical consequences of noun incorporation, we refer the reader to Chung and Ladusaw 2020 and references therein.

3.1.5 Polysynthesis

The label POLYSYNTHESIS is usually used for languages with very high morpheme-to-word ratios, in particular forming quite complex verbs that can stand alone as sentences, the components of which would be expressed as independent words in other languages. Such sentence-like verbs are one of the reasons polysynthetic languages have received so much attention in the literature; the Morphology—should it be uniquely responsible for generating words—must do quite a lot of work in polysynthetic languages, work that is usually the job of the Syntax.

Many Indigenous languages of the Americas are said to be polysynthetic, as are many other languages areally clustered in, for example, northern Australia and Papua New Guinea (Fortescue et al., 2017). Consider, for example, the Tlingit (Na-Dene) example in (23), which is a verb and also a sentence:

- (23) [ʔiχ^w.sì.'tút']
 i-wu-x-s-i-dut'-H
 2SG.OBJ-PFV-1SG.SBJ-CAUS-STV-hiccup-VAR
 'I have made you hiccup.' (Tlingit, Crippen 2019, p. 1)

Tlingit regularly forms these kinds of large verbs, and indeed, the language as a whole is traditionally termed polysynthetic.

It is important to note that the term 'polysynthesis' has been used in many different ways in the literature (see e.g. Fortescue et al. 2017; Barrie and Mathieu 2019; Johns 2023, and references therein). Crippen (2019, §1.1.2) notes that the label 'polysynthetic' in practice “seems to mean little more than ‘verbs have lots of morphemes’” and that this classification can thus be misleading in implicating some unifying property—polysynthetic languages are not a monolith. The label is also sometimes associated with certain specific properties beyond high morpheme-to-word ratios, such as the optionality of overt nominal arguments, incorporation of nominals into the verb (see §3.1.4), and free word order.

Are polysynthetic languages different in some fundamental way from non-polysynthetic languages? E.g., is there a polysynthesis “parameter” (Baker,

1996)? Or can polysynthesis be captured without any special mechanisms, which is what Crippen (2019) argues for Tlingit? The consensus view today is that polysynthetic languages do *not* require any unique grammatical mechanisms: these languages utilize the exact same sorts of processes found in non-polysynthetic languages—affixation, compounding, and noun incorporation (see §3.1.2-4). The Kiowa (Kiowa-Tanoan) example in (24) illustrates all of these processes—the verb is inflected for agreement, an argument of the verb (‘make’) is incorporated, and this incorporated argument is itself a compound (deer-hide). (Note that *hóldà* ‘shirt’ is lexically plural.)

- (24) Hóldà gát-tháp-kàui-ǎúmé.
 shirt 1SG>3PL-deer-hide-make.PFV
 ‘I made a shirt with/out of buckskin.’
 (Kiowa, McKenzie 2022, p. 516)

The idea is that so-called polysynthetic languages differ in degree of word-formation, but not kind.

Non-lexicalist approaches must syntactically contend with both the regularities and idiosyncracies of word-formation in polysynthetic languages. For Kiowa, (24), McKenzie (2022) specifically puts forward a syntactic analysis of “polysynthesis”, proposing a mediating relation—represented as a syntactic head—to select incorporated nouns and thematically relate them to the verb. This proposal can in turn more generally address the weak compositionality of such structures, as well as that found in compounding, in the syntactic component of the grammar. In §3.2.3, we return briefly to the connection between polysynthesis and syntax, in the context of the nature of wordhood.

3.1.6 Word size and morphological theory

§2 differentiated theories based on whether there is a wholly non-syntactic morphological module that is responsible for building words, as in lexicalist theories, or not, as in non-lexicalist theories. The great variation in word complexity discussed above is highly relevant to this discussion.

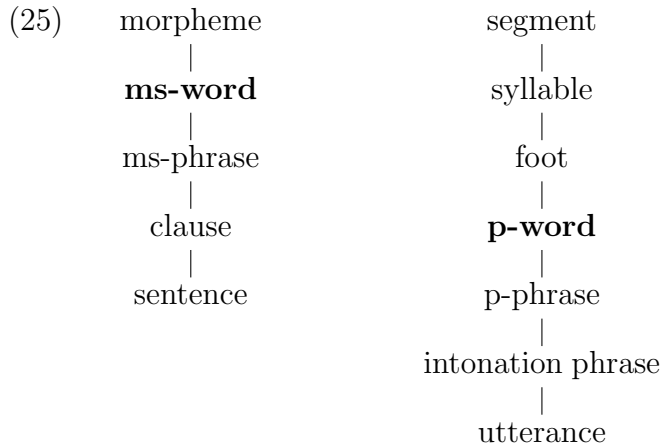
According to lexicalist theories (see §2.2), isolating languages have very little going on in the morphological component of the grammar because of the near-absence of complex words, and (potentially) much more going on in the syntactic component of the grammar. Polysynthetic languages are the reverse, as they must have heavy morphological modules, and they may have comparatively light syntactic modules. Synthetic languages have more balanced morphological and syntactic modules, and it is these kinds of languages that are the bread-and-butter of lexicalist related approaches.

In non-lexicalist theories (see §2.3), the trade-offs across the grammar are of a different nature, as all linguistic expressions, regardless of size, originate in the Syntax; the Syntax is therefore *always* in the business of manipulating morpheme/feature-sized elements, in isolating and polysynthetic languages alike (and everything in between). In a post-syntactic approach like DM, there may be significant differences across languages in terms of the heaviness of the post-syntactic morphological module. Consider, for example, Arregi and Nevins’s (2012) approach to the Basque (isolate) auxiliary, which requires a particularly intricate and multi-faceted post-syntactic module. In purely syntactic approaches to morphology, all morphological *and* syntactic action is in the Syntax, though different aspects of syntactic computation may be much more utilized in some languages than others. Consider, for example, Shlonsky’s (2023) syntactic approach to Tamazight and Hebrew (both Afro-Asiatic) inflection, which requires complex interactions of syntactic movements with (morpho)syntactic licensing.

3.2 But wait, what is a word? Criteria and issues

Recall that the primary differentiator among theories of the morphology/syntax interface is whether morphology and syntax meet at the level of the word (lexicalism) or not (non-lexicalism); see §2.1.1. But what even is a word? Identifying words and word boundaries is an implicit or explicit component of nearly all linguistic work, across subfields and theoretical orientations. Yet the basis for this demarcation remains a matter of ongoing debate, and the notion of what constitutes a “word” is far from straightforward.

One area of robust consensus revolves around the idea that to solve the wordhood problem, we (minimally!) need to distinguish between the unit that (morpho)syntax treats as a word—so-called grammatical words or morphosyntactic words, henceforth referred to as MS-WORDS—and the unit that phonology treats as a word—so-called phonological words or prosodic words, henceforth referred to as P-WORDS. Intuitively, these two notions of wordhood can be understood as picking out a particular type of morphosyntactic or phonological/prosodic constituent, respectively, each occupying a rung of the relevant hierarchy, morphosyntactic or phonological/prosodic (see, e.g., Selkirk 1981; Dixon and Aikhenvald 2003):



An ms-word is a constituent that (i) contains one or more morphemes (or, in a non-morpheme-based theory, stem and/or feature) and (ii) is contained in a morphosyntactic phrase (MS-PHRASE). A p-word is a constituent that (i) contains one or more syllables (themselves composed of segments), grouped into one or more feet, and (ii) is contained in a phonological phrase (P-PHRASE). The motivations for positing these different levels—including ms-word and p-word—ultimately boil down to showing that rules or generalizations of grammar need to refer to, or apply uniquely at, a particular level.

The separation of ms-words from p-words affords a lot of empirical and theoretical traction (though it is not a panacea). In the remainder of this subsection, we discuss common criteria for defining ms-words and p-words (§3.2.1-2), and then turn to empirical profiles of mixing and matching across these notions of wordhood (§3.2.3).

3.2.1 Morphosyntactic/grammatical criteria for wordhood

There are several criteria very commonly taken to identify ms-words, and more generally, subword elements that are bound morphosyntactically (see, e.g., Dixon and Aikhenvald 2003; Julien 2007; and references therein), including cohesiveness, fixed order, selectivity, and domainhood for morphological operations. We go through these criteria in turn.

The criterion of cohesiveness is about (non-)interruptability of ms-words:

- (26) *Cohesiveness*: While ms-words can be separated from each other, proper subparts of an ms-word cannot.

Consider a Bosnian (Slavic) sentence, (27). Ordering is flexible: the verb and object need not be adjacent, the demonstrative and noun can separate, etc.

- (27) a. Po-sjek-l-i **su** ov-o drv-o.
 PFX-cut-PTCP-PL be.3PL.PRES this-N.SG.ACC tree-N.SG.ACC
 ‘They cut this tree.’
 b. Ovo drvo **su** posjekli.
 c. Ovo **su** drvo posjekli.
 d. Ovo **su** posjekli drvo. (Bosnian)

Important for this illustration is that the bolded auxiliary *su* in (27) is a second-position element, requiring something to its left. Even though very phonologically light things (which are themselves not phonologically independent) can serve as first position ahead of *su* (Talić, 2018), nevertheless, *su* can never appear inside the verbal complex, leaning to its left on some subpart of this complex:

- (28) a. *po- **su** sjek-l-i ovo drvo.
 b. *po-sjek **su** -l -i ovo drvo.
 c. *po-sjek-l- **su** -i ovo drvo. (Bosnian, cf. (27a))

This restriction on *su* can be understood if *po-sjek-l-i* is an ms-word, and thus cohesive—*su* cannot interrupt this constituent.

A second common criterion for ms-wordhood is the rigidity of the ordering of sub-elements:

- (29) *Fixed order:* While ms-words can be reordered with respect to each other, the proper subparts of an ms-word cannot.

As we saw in (27), Bosnian has relatively flexible word order. However, even in such a language, the order of word-internal elements is fixed. For example, the verb ms-word that we saw above can only have one order of its internal elements:

- (30) a. po- sjek -l -i
 PFX- cut -PTCP -PL
 b. *sjek-l-i-po
 c. *po-sjek-i-l
 d. *l-po-sjek-i
 e. *i-po-sjek-l (Bosnian)

In contrast, ms-words themselves may be characterized by some freedom of ordering. Indeed, as implied by (29), ms-words are often posited to be the smallest syntactically-mobile unit. When order *can* change inside an ms-word, the word usually has different meanings depending on the ordering (e.g.,

the non-equivalence *workshop* and *shopwork*, or *non-pre-existent* and *pre-non-existent*). (See, e.g., Rice 2000 on semantic scope and morpheme order.)

Related to both cohesiveness and fixed order is that subword elements that are functional in nature (e.g., affixes) are generally picky to a higher degree than independent words.

- (31) *Selectivity*: Ms-words are less selective with respect to what they combine with compared to affixes (or other bound functional elements).

For example, the comparative suffix *-er* can only combine with certain stems, generally short and frequent ones (*silli-er*, **fantastic-er*) (Graziano-King, 1999). The free word *more*, which also has a comparative meaning, can combine with a much wider set of elements (*more fantastic*; *more sinks*; *more quickly*; *sleep more*), even those adjectives that usually combine with *-er*, given the right pragmatic context. Note that while words can impose selectional requirements on a complement (e.g., *devour* specifically selects for an NP), they don't (usually) impose requirements on the particular lexical item they combine with.

Within the tightly bound domain picked out by cohesiveness, fixed order, and selectivity, it is common to find operations that are restricted to applying within this domain. For example, when a given meaning/function corresponds to multiple potential exponents (see §4.1 on suppletive allomorphy), the conditioning factors for choosing among the exponents are found within the ms-word, at least in the vast majority of cases. As another example, recent work on Ethiosemitic verbs (Hewett and Kramer, 2026) proposes a constraint requiring the uniqueness of agreement within an ms-word to explain the (dis)appearance of certain agreement marking across different contexts. Such domain-bound phenomena support the delineation of an ms-word unit.

As is well-documented in the literature (see, e.g., Marantz 1997; Haspelmath 2011; Bruening 2018; Tallman 2020), none of the criteria discussed above are fully sufficient to diagnose ms-wordhood, nor are they unique to the ms-word level. For example, words in phrases are often in fixed orders with respect to each other and may be inseparable (e.g., *very tiny*), words can be picky about the lexical items they combine with (e.g., as in so-called particle verbs like *throw out* or *pick up*), and morpheme ordering within a word may be variable without a concomitant change in meaning. In addition, even in a single language the above criteria can pick out conflicting units, not all converging on the same size constituent (Zingler, 2020). These challenges to defining ms-words lead some to reject the notion of ms-wordhood entirely. We return in §3.3 to how the above criteria relate to ms-wordhood in different theories of word-formation, including the possibility that ms-wordhood is an

illusion, i.e., there are no ms-words (§3.3.4).

Although there are problems in defining ms-wordhood, we maintain the at least descriptive usefulness of the ms-word to pick out a particularly closely-bound morphosyntactic unit, which may be a matter of degree rather than categorical, but without which we have no hope of empirically differentiating morphology and syntax at all.

3.2.2 Phonological/prosodic criteria for wordhood

A different type of closely-bound unit that can be identified as a kind of word is the p-word. Two types of evidence are typically marshaled to identify p-words (Selkirk, 1984; Nespor and Vogel, 1986; Elordieta, 2014): (i) the bounding of phonological/prosodic processes or generalizations to the p-word, and (ii) the necessity of referencing p-word edges in the formulation of specific phonological rules or prosodic patterns.

Let's consider bounding first. Phonotactic constraints that are bounded to the p-word include minimal weight/size requirements, e.g., p-words in many languages must reach a certain number of moras (a measurement of rhythmic weight), e.g., in Mono (Adamawa-Ubangi; Olson 2003). Phonological processes, e.g. assimilation, vowel harmony, tone spreading, and tone shift, can also be p-word-bound. As a segmental example, in Movima (language isolate of Bolivia; Haude 2006, p. 34-35), nasal assimilation is obligatory, with /n/ assimilating in place to any following consonant within the same p-word; but, nasal assimilation crucially does not take place across p-words, i.e., for word-final /n/. As a tonal example, in Silozi (Bantu), tone shift processes are strictly bound to the p-word domain (Bickmore, 2024, p. 219–221).

Another way p-words reveal themselves is through edge phenomena: there are phonotactic constraints, phonological rules, and prosodic patterns that specifically reference the edge of a p-word. For example, in a wide range of languages, word-initial onsets and word-final codas behave differently than the onsets/codas of other syllables; see Flack 2009 for an overview. Prosodic features also demarcate p-words, e.g., stress assignment is generally sensitive to prosodic word edges (see Elordieta 2014 and references therein).

Together, these phenomena demonstrate the utility of the p-word as a particular level of prosodic structure that the grammar can refer to. Sometimes, though, different criteria for diagnosing p-word boundaries can conflict within a single language, showing—just like with ms-word domains—that it can be difficult to definitively identify the p-word, at least as a unique level (see Zingler 2020 and references therein). In Turkish, for example, as discussed by a number of works (Kornfilt, 1996; Newell, 2008; Fenger, 2020),

	ms-free	ms-bound
p-free	canonical word	non-cohering affix
p-bound	simple clitic	canonical affix

Table 3: Typology of boundness

both vowel harmony and stress assignment identify p-word-like constituents. While these criteria usually converge on the same strings, they come apart in certain tense/aspect/mood configurations. Why that might be—and its implications for morphological/syntactic theory—is best understood in the context of the mismatches we turn to next.

3.2.3 Mix and match

To a large extent, the criteria for ms-wordhood, §3.2.1, and p-wordhood, §3.2.2, align with each other, creating the impression that they are one and the same “word”. However, mismatches between these two units also robustly occur across languages, showing it’s essential to differentiate them.

To explore this, we need to refine our definition of the bound vs. free distinction introduced in §2.1.3. Elements that are MS-BOUND are identified by the criteria in §3.2.1 (cohesiveness, fixed order, selectivity, domain of operations) as necessarily being internal to a (complex) ms-word; MS-FREE elements are the converse. Elements that are P-BOUND are identified by the criteria in §3.2.2 (phonotactic, phonological, and prosodic behaviors) as necessarily being internal to a (complex) p-word; P-FREE elements are the converse. Combining the ms-free vs. ms-bound distinction with the p-free vs. p-bound distinction yields a four-way typology, summarized in Table 3. When an element is ms-free and p-free, this is a canonical (monomorphemic) word, like English *cat*. When an element is ms-bound and p-bound, this is a canonical (cohering) affix, like English plural *-s*. The mismatched cells are dealt with below.

The term (SIMPLE) CLITIC²¹ commonly refers to an element that acts like it is a regular, independent morphosyntactic unit (ms-free), but phonologically it is bound; being p-bound in this case may either mean the element looks/acts like a canonical affix phonologically or that it otherwise is phonologically de-

²¹A division between “simple clitics” and “special clitics” was first made by Zwicky (1977): special clitics, in contrast to simple clitics, have a surface distribution that is different from where we’d expect a related free form to occur. The term “clitic” also has a specialized usage to refer to (what is analyzed as) a reduced pronoun (see, e.g., Kramer 2014). It is an open question whether these are all the same types of elements.

ficient, e.g., accentless words and “leaners” (Nevis et al., 1994). Compare, for example, the English plural *-s* with the English possessive *-s*: although both are p-bound, possessive *-s* does not act like an actual affix, since it doesn’t have any morphosyntactic relationship with the word it phonologically is bound to (e.g., *the dean who I had a Zoom call with’s cat*) and arguably distributes like a (free) determiner (Radford, 1992); this would make possessive *-s* a clitic. In contrast, plural *-s* only combines with nouns; it is a canonical affix. The typical analysis of simple clitics, across theories, is that they are syntactic heads (or even phrases) but phonologically special, for example, requiring prosodic incorporation, and potentially belonging to a unique level of prosodic structure, the Clitic Group (Selkirk, 1981; Elfner, 2012).

A final possibility is that an element could be ms-bound but p-free, i.e., something that seems to be in a tight morphosyntactic relationship with another element, but nevertheless acts as its own phonological domain; this is essentially the inverse of a simple clitic. Although there is no term in widespread use to refer to this configuration of ms- and p-boundness, such elements were termed NON-COHERING AFFIXES by Dixon (1977), and this term has stuck around, particularly to differentiate subtypes of Germanic affixes. In Dutch, for example, all prefixes are non-cohering (p-free), and suffixes may be cohering (p-bound) or non-cohering, forming their own p-word (Booij, 1995b).

An interesting question arises for polysynthetic languages (see §3.1.5): what notion of wordhood is being referred to, p-wordhood, ms-wordhood, or both? The answer varies depending on the language. In Inuit (Inuit-Yupik-Unangan), for example, it has been argued that the elements in a polysynthetic word are morphosyntactically differentiable, i.e., they are (for the most part) ms-free; but, these ms-free elements are p-bound and join together to form a very large p-word. Specifically, Compton and Pittman (2010) propose that the syntactic CP and DP levels spell out as p-words, giving the appearance of polysynthesis. Conversely, in Chintang (Kiranti), it looks (more like) there is a large tightly-bound morphosyntactic domain, at least in terms of being cohesive, but several phonological domains can be found in this larger morphosyntactic domain; see Bickel and Zúñiga 2017, which also discusses the overall difficulty with identifying consistent domains within polysynthetic languages. This variety confirms Crippen’s (2019) conclusion that polysynthetic languages are not a monolith.

The mismatches represented in Table 3, alongside the challenges in defining consistent ms-word and p-word domains even within a single language (see the ends of §3.2.1 and §3.2.2), pose a major puzzle for linguistic theory. One response is to wholesale deny the utility of “word” as a unit of linguistic analysis, in either the ms-word sense, the p-word sense, or both. Many have been

dissatisfied with this move, however, for the simple reason that morphosyntactic and phonological notions of wordhood do in many cases align or at least stand in a systematic relationship (see, e.g., Match Theory, Selkirk 2011).

An important empirical consideration in theoretically exploring ms-wordhood and p-wordhood is CYCLICITY. Defined very generally, cyclicity is when a particular computation applies (or at least appears to apply) more than once, over increasingly-sized strings/units. Cyclicity appears in (morpho)phonology (see Bermúdez-Otero 2011 and references therein) and in (morpho)syntax (see Freidin 2017 and references therein). The question thus arises as to whether (morpho)phonological cycles and p-wordhood have any relation to syntactic cycles and ms-wordhood. Lexicalist approaches necessarily hold that the two notions of cyclicity/wordhood are independent, one defined in the morphological component (e.g., strata; Kiparsky 2000; Bermúdez-Otero 2017), and the other defined in the syntactic component (e.g., phases; Chomsky 2001a). On the other hand, an active strand of research within non-lexicalist approaches explores the idea that in fact these cycles are one and the same, originating in the Syntax with syntactic cycles, and in turn giving rise to both ms-wordhood and p-wordhood (and their mismatches); see, for example, the approaches pursued in Newell 2008, Fenger 2020, Talić 2019, and Fenger et al. 2025.

Moving forward in this Element, we put p-wordhood aside, and we explore instead ms-wordhood. It is crucial to note here that not all theories are created equal in terms of their reliance on ms-words (or some related designator of what is syntax vs. morphology). In general, the existence of ms-words is crucial for lexicalist theories, because the word level defines the interface between morphology and syntax. Although non-lexicalist theories frequently make use of the notion of ms-word, they are not constrained by it in the same way, because such approaches take all types of linguistic expressions to originate in the syntactic component of the grammar.

3.3 How (and why) are morphosyntactic words formed?

This section reviews various explanations that have been proposed for the formation of ms-words, which we henceforth refer to as just “words” (since we put p-words aside). We begin briefly with lexicalist approaches, before turning more extensively to non-lexicalist approaches.

In lexicalist and non-lexicalist approaches alike, it is often assumed that words correspond to syntactic heads (X^0 s), though this is a more central and indispensable assumption of lexicalism, for which words constitute the interface between morphology and syntax (see §2.2). In lexicalist approaches, the word-building operations of the Morphology might resemble those of the Syntax,

or may be quite different (see §2.1.3). These operations, alongside language-specific paradigms, lexical items, etc., determine the size, form, and content of words in a given language. Note that within lexicalist approaches, the apparent correspondences between word-internal and phrase-level ordering/structure—Baker’s (1985) Mirror Principle, noted at various points throughout this Element—must either be spurious or receive an explanation that does not stem from a unification of morphology and syntax. For example, it has been argued that apparent Mirror Principle effects arise from semantic relatedness to the root and forces of historical change (Bybee, 1985).

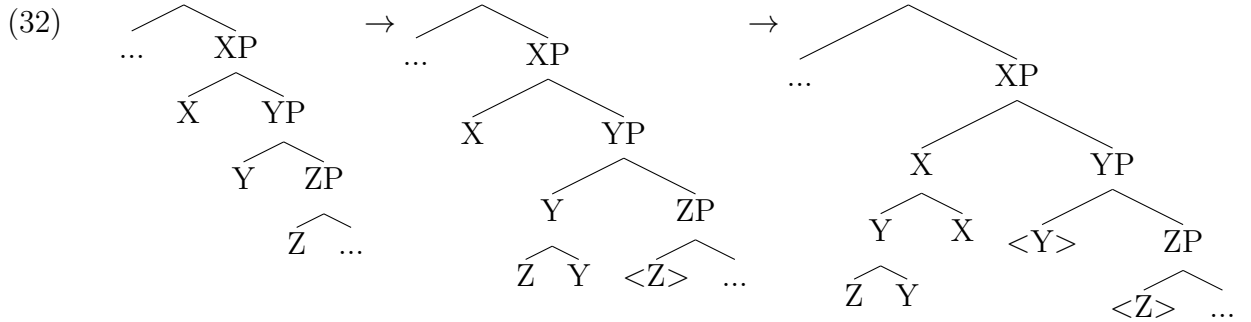
For a number of reasons, many contemporary theories of the morphology/syntax interface have moved away from lexicalism and towards a position where syntax *is* involved in word-formation, or at least underlies it (see §2.3). Words often appear to contain complex internal structure, complexity that would correspond to multi-word syntactic phrases in other languages (or even within the same language), with systematic parallels to syntactic and semantic structure (again, as expressed by the Mirror Principle). Further, all languages *need* at least a syntactic component, whereas the need for an independent morphological component is limited in at least some languages, most starkly in isolating languages (see §3.1).

For the remainder of this section, we discuss the formal mechanisms of word formation invoked by non-lexicalist approaches. The discussion is organized by the syntactic configurations proposed to constitute words: syntactic heads (§3.3.1), sequences or spans of heads (§3.3.2), and post-syntactic amalgams of heads (§3.3.3). We also discuss the possibility that (at least some) “words” are not a morphosyntactic constituent nor in a particularly-defined configuration at all (§3.3.4).

3.3.1 Words as syntactically complex heads

For non-lexicalist theories, complex word formation is often attributed to syntactic head adjunction, via either upward head movement or head lowering. Here we provide a brief overview of head adjunction as a mechanism that builds complex words; for more detailed discussions we refer the reader to Harley 2013 and Dékány 2018.

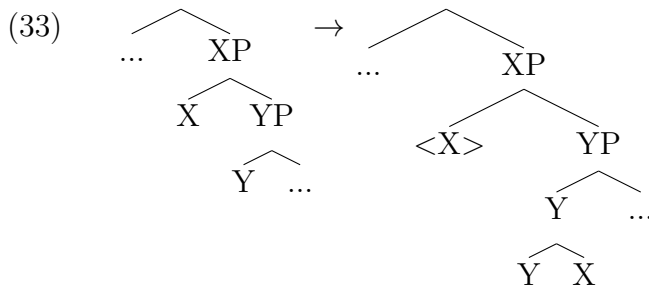
HEAD MOVEMENT is an operation that raises the head of a phrase to adjoin to the head of a syntactically higher phrase, as shown in (32) with Z first raising to adjoin to Y, and then the newly-formed head complex in Y raising to X.



Head movement is generally presumed to obey the Head Movement Constraint (Travis, 1984), which requires such movement to be strictly local; accordingly, in (32), Z can raise to Y , and Y can raise to X , but Z could not raise directly to X . Head movement is also generally assumed to “snowball”, with all heads in the path of head movement being brought along with further movement. On this type of approach, potential morphological complexity within words scales with the extent of head movement along a syntactic spine.

As can be seen in (32), local snowballing head movement necessarily recapitulates the hierarchical ordering of the underlying syntax. Thus, the most-embedded head (before head movement) is the most-embedded head in the complex head adjunction structure, the next-most-embedded head (before head movement) is the next-most-embedded head in the complex head adjunction structure, and so on. It is precisely this that automatically gives rise to Mirror Principle effects.

Alongside upward head movement, some theories include a mechanism of head lowering that allows heads to move *downwards* to adjoin to a lower head, as illustrated in (33).



As an operation, this goes back to Chomsky’s (1957) Auxiliary Transformation (later referred to as Affix Hopping), reframed in structural terms here. Beginning with Pollock (1989), it is widely assumed that verbs in some languages, such as French, combine with finite tense and agreement inflection by head moving upward overtly to Infl/T, while in other languages, like English,

the finite verb remains *in situ* and can only combine with finite inflection via some kind of downward head adjunction.

A final syntactic possibility for building words is that complex heads are built directly, via first merge (not head movement), using slightly modified versions of the usual operations of syntax. This is the approach of *non-integrated* syntactic approaches to morphology (see §2.3; e.g., Lieber 1992; Bruening 2017). These complex heads then behave essentially the same as simplex heads for the purposes of further structure building above the head level. Of course, an approach might also mix head movement and direct generation of complex heads, for different cases, as is sometimes the case in DM.

Why are complex heads built in the first place? If complex heads are built directly (i.e., as in non-integrated approaches), word-formation is largely about the selectional properties of particular types of elements in a language. If complex heads are instead derived by syntactic movement (i.e., as in integrated approaches), then the answer depends in part on the freedom with which movement takes place in a theory. The operation “Move- α ” of Government and Binding Theory (Chomsky, 1981) posited that movement applied freely, i.e., did not need to be triggered. It was within this context that the Stray Affix Filter (Lasnik, 1981) was born, as a well-formedness condition requiring bound morphemes to have a host. Under this type of approach, the boundness of an affix in T (for example) does not trigger V-to-T movement in French, but rather this movement simply takes place as a result of movement freely applying. In English, when verb movement is independently ruled out, a *repair*—DO-support—provides an affix in T with an appropriate host. Bjorkman (2011) makes a similar claim instead for auxiliary forms of BE.

In contemporary Minimalist approaches, it is instead assumed that movement must always be triggered. Therefore, if syntactic movement gives rise to words, then some feature or structural configuration must trigger that movement.²² In some approaches, some heads are designated as being dependent or bound, essentially a version of the so-called “EPP” features that exist to trigger phrasal movement (see, e.g., Arregi and Pietraszko 2018). As an alternative, Richards’s (2016) Contiguity Theory attributes head movement to the need for heads in certain dependencies with each other to be prosodically contiguous (i.e., within the same prosodic domain).

All the approaches discussed above in this section hold that a word, in the usual case, corresponds to an X^0 node in the Syntax. Certain aspects of ms-

²²In lexicalist versions of Minimalism (e.g., Chomsky 1993), where verbs were assumed to enter the Syntax fully inflected, head movement did not build inflected verbs but instead provided a configuration in which tense and agreement features on a verb could be checked.

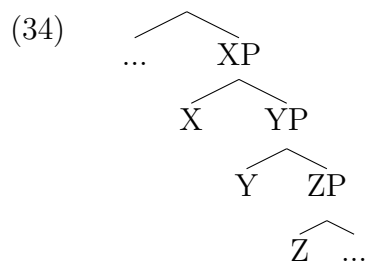
wordhood discussed in §3.2.1 are naturally accounted for if words are complex heads: another word or phrase cannot intervene inside a complex head; fixed order within the word is due to the strict locality of head movement; and selectivity may arise in this particularly tight structural configuration. The X^0 node can also be referred to as the domain for processes that appear to be bound to the *ms-word*, like suppletive allomorphy (see §4.1).

However, the field has largely moved away from traditional syntactic head movement, due to both empirical concerns—height and morphological complexity do not always correlate—and theoretical concerns—syntactic movement should be *upward*, and should not violate the Extension Condition (see Chomsky 1995a, 2001b). Some approaches aim to address (some of) these concerns while maintaining syntactic head movement, e.g., Arregi and Pietraszko (2018) capitalize on the possibility of *covert* movement (Chomsky, 1991) to explain why a head can be pronounced in a lower position than it syntactically moved to. We explore some further alternatives in the coming sections.

3.3.2 Words as sequences/spans of heads

In the last two decades, there have been several proposals that seek to map words onto sequences of heads, rather than single heads (whether simple or complex). Among the first such proposals appeared in Brody 2000 with the development of Mirror Theory. Brody proposes that what looks like both upward and downward head movement arises due to the labeling properties of heads in a functional spine: according to the Bare Phrase Structure assumption that a head and its maximal projection bear the same label (Chomsky, 1995a), there is no distinction between the phrasal complement (XP) of a head H and the head of that complement (X), and so a sequence of heads will linearize into a single word unless prevented from doing so, e.g., by occurring in a specifier position; this word can appear at the syntactic height of any of the implicated heads. (See also Adger 2013.)

Building in some ways on Brody’s approach, Svenonius (2016) invokes the unit *span*, a sequence of heads in a head-complement relation. For example, X, Y, and Z in (34) form a span:



(e.g., Kramer 2010). What motivates M-Merger, Lowering, and Local Dislocation may be post-syntactic well-formedness requirements akin to the Stray Affix Filter (e.g., Yuan 2025), or language-specific post-syntactic operations tied to specific lexical items or syntactic categories.

Refining the tool(s) of post-syntactic word-formation, Harizanov and Gribanova (2019) argue that what is called “head movement” seems to encompass two distinct classes of phenomena: those that displace lexical words (and arguably have semantic consequences), potentially long distances, and those that form morphologically complex words out of contiguous heads (without semantic consequences). They propose that the strictly-local, word-forming kind of “head movement” is post-syntactic, with both a raising and lowering counterpart, conceptualized in a unified way as head AMALGAMATION. Amalgamation is motivated by the presence of a morphological selection feature, which in turn may require a head to lower or raise in the post-syntax.

What all of these approaches share is that words are built from the bundling of multiple syntactic heads into a unit, as part of the post-syntactic morphology. Depending on the particular formulation of the post-syntactic mechanism, various aspects of the properties of ms-words discussed in §3.2.1 may be accounted for. Importantly, note that if *all* word-formation is post-syntactic, this means that there is essentially no syntactic notion of word, only a post-syntactic one. Alternatively, if both syntactic and post-syntactic word-formation is possible (e.g., raising is a syntactic operation, but Lowering is post-syntactic), then there is the possibility for a third notion of wordhood, differentiating syntactic words (complex heads in the Syntax) from post-syntactic words (complexes that arise in the post-syntax); if this is correct, we expect different characteristics to potentially emerge for these different types of words.

3.3.4 No morphosyntactic words

A final possibility is to deny that morphosyntactic words actually exist at all, and rather, to say that our perception of ms-words as real is due to frequent/conventionalized collocation and/or status as a p-word (§3.2.2). Julien (2002), for example, proposes that while complex words may be formed syntactically via head movement or base generation (à la §3.3.1), wordhood is often just “apparent”, resulting from frequently-co-occurring sequences of heads (morphemes) that don’t have other material intervening between them. The framework of Nanosyntax (see §2.3) also generally falls in this camp, with “words” defined by the phonological characteristics of morphemes, alone or in combination, and not by syntactic or post-syntactic structure.

If there are no morphosyntactic words, this opens the door for the Syntax

to play a more radical role in getting affixes in the right order. For example, it has been proposed that morpheme order results from phrasal movement to specifier positions (see, e.g., Koopman 2017; Myler 2017; Shlonsky 2023). Of course, to the extent that *ms*-words form a domain for certain processes (§3.2.1), something else apart from actual wordhood must explain this.

3.4 Section summary

In this section, we explored the empirical and theoretical dimensions of what constitutes a “word” in linguistic analysis, highlighting variation in word complexity across languages (§3.1), attempting to define wordhood (§3.2), and exploring the mechanisms posited for creating words (§3.3).

4 Complexities of form-meaning mappings

A crucial aspect of morphological research involves systematically decomposing words into their component pieces or processes and matching those phonological exponents to particular meanings/functions. This task is a necessary step in *any* attempt to model the morphology/syntax interface (§2), and arises no matter the role of the notion of “word” in a given theory (§3). This section focuses on the different kinds of relationships that can hold between pieces and processes (exponents) on the one hand and the morphosyntactic building blocks of meaning (morphosyntactic features and roots) on the other; following the convention in the field, we refer to these relationships as FORM-MEANING MAPPINGS.

What types of form-meaning mappings are there, from an empirical, descriptive perspective? Let’s start with variation on the form side, holding meaning/function constant. For expressing one particular meaning/function, there might be one isolable phonological exponent, in a simple, ONE-TO-ONE form-meaning mapping where the exponent never varies. The root *cat*, for example, is always /kæt/ in English, and progressive marking is always *ing*, /ɪŋ/. Alternatively, for expressing one particular meaning/function, an exponent might be one of several competing options, a situation known as (suppletive) ALLOMORPHY. When a noun in English is pluralized, there are a number of competing exponents of the plural, e.g., [-z] for *dog*, [-ən] for *ox*, and a zero exponent for *sheep*, *aircraft* (etc.). A final possibility is that a particular meaning/function could be expressed via multiple co-occurring (*non*-competing) exponents, giving rise to MULTIPLE EXPONENCE. For example, the noun *k’al* ‘word’ in Amharic (Semitic) can have two plural suffixes, while having a single

plural interpretation: *k'al-at-otftf* ‘word-PL-PL’ (Kramer, 2016a, 530). Both allomorphy and multiple exponence involve some kind of *abundance* on the form side of a form-meaning mapping.

Let’s turn now to the meaning/function side, holding form constant. As above, a particular meaning/function could have a one-to-one mapping to an isolable phonological exponent. Alternatively, a set of grammatically-related but distinct (non-co-occurring) meanings/functions could be expressed by one exponent, a situation known as *SYNCRETISM*. For example, for regular verbs in English, the past tense exponent is the same as the passive participial exponent, specifically, the *-ed* suffix (*I rinsed the vegetables; the vegetables were rinsed*). As a different kind of abundance on the meaning/function side, a set of isolable meanings/functions could jointly (i.e., specifically when co-occurring) be expressed by a single exponent, giving rise to a *PORTMANTEAU*. For example, the exponent *am* in English encompasses a set of morphosyntactic meanings/functions: 1st person, singular, and present tense, combined with the copula *be*.

Two final kinds of basic form-meaning mapping involve *absence* rather than abundance. On the one hand, a meaning/function might not be mappable onto a form at all, resulting in a *MORPHOLOGICAL GAP*. For example, the English verb *stride* cannot be put into past participial form for many speakers (*stridden* is often rejected). Conversely, a phonological exponent might not be mappable onto any meaning/function, resulting in an *EMPTY MORPH* that seems to simply take up morphological space. For example, verbs in Romance languages have a decomposable “theme vowel” that is an integral part of the inflectional system yet is not (apparently) meaning- or function-bearing.

Mixing and matching among the above types of abundance and absence phenomena is possible, too, giving rise to yet-more-complex form-meaning mappings. For example, an empty morph might be involved in allomorphic alternations, as in the case of Romance theme vowels. A syncretic exponent could itself be a portmanteau, as with the inflected copula *are* in English. And so on. To the best of our knowledge, all possible combinations of the above phenomena are both logically-possible and attested.

While the different types of form-meaning mappings described above might seem like they define straightforward, separate buckets to sort phenomena into, it is crucial to recognize the distinction between descriptive (empirical) buckets on the one hand and theoretical buckets—actual analyses—on the other hand. Theories generally have fewer theoretical buckets than there are empirical buckets. And even deciding which empirical bucket a given case belongs in is unavoidably a matter of analysis to at least some degree. Throughout the section, we highlight and discuss many examples of phenomena whose type

of form-meaning mapping depends on analysis, as this is one of the most important ways in which form-meaning mappings present complexities for morphosyntactic research.

This section is organized into empirically-based buckets, starting with abundance on the form side of the mapping, with allomorphy (§4.1) and multiple exponence (§4.2), then turning to abundance on the meaning/function side of the mapping, with syncretism (§4.3) and then portmanteaux (§4.4), and finally discussing briefly the mappings involving absence, morphological gaps (§4.5.1) and empty morphs (§4.5.2). We conclude the section with a table summarizing the relation of these phenomena to certain theories (§4.6). As we will see, fundamental differences emerge across theories in how imperfect form-meaning mappings are encoded or derived. Generally-speaking, realizational theories predict more kinds of “imperfect” (non-one-to-one) mappings to be possible than incremental theories do (see §2.1.4), with some incremental theories allowing *only* one-to-one mapping.

4.1 Suppletive allomorphy

This section considers the empirical situation when there is an abundance of exponents in competition for expressing a given meaning/function. It’s important to first draw a distinction between the kind of allomorphy we’re interested in here and other uses of the term “allomorphy.” Allomorphy, as a general term, has been used to refer to *any* type of shape alternation in/among exponents, based on their context. Usually, though, this term is reserved for alternations that can’t be explained just by the regular phonology of a language,²⁴ which is in line with our usage here.

Even putting aside regular phonology, allomorphy picks out a big empirical space, which would include both types of alternations seen in (36): in (36a), a wholly different form of the root appears in the context of the comparative *-er* suffix; in (36b), just the final consonant of the root is affected in the presence of the plural suffix.

- | | | |
|------|-----------------------------------|------------------|
| (36) | Types of (true) allomorphy | |
| | a. good vs. bett-er | (suppletive) |
| | b. leaf vs. leaves | (non-suppletive) |

The type of alteration seen in (36a) is SUPPLETIVE ALLOMORPHY, where the exponents are wholly different. The type of alternation in (36b) is NON-

²⁴Although sometimes even alternations explained by regular phonology are included under the umbrella of “allomorphy,” specifically, as *surface* allomorphy.

SUPPLETIVE ALLOMORPHY, involving alternations of subparts of an exponent that are plausibly phonological in nature but not attributable to regular phonology.²⁵ The nature of non-suppletive allomorphy situates it more centrally at the morphology/phonology interface, and so we put it aside here.²⁶ Going forward, we exclusively deal with suppletive allomorphy. Note that we use this as a general term to cover replacive allomorphy even when it is an affix that has a distinct form in a particular context. (Some use this term only to describe root/stem allomorphy like that in (36a).) And of course, it is not always clear whether to treat a given case of allomorphy as suppletive or non-suppletive; see discussion in e.g. Kalin 2022.

Empirically, a variety of factors can impact the choice among suppletive exponents for expressing a particular meaning/function, including a phonological aspect of the environment, as in (37a), a morphosyntactic aspect of the environment, as in (37b), or a lexical aspect of the environment, as in (37c).

(37) Examples of suppletive allomorphy

- a. Moroccan Arabic (Semitic; Paster 2006, 56, who cites Harrell 1962, 136)

xt ^ʕ a- h	vs.	ktab- u
error-3SG.M		book-3SG.M
‘his error’		‘his book’
- b. Adiyaman Kurmanji (Indo-Iranian; Kalin and Atlamaz 2018, 93)

kel- i	vs.	kel-on- d
boil-PAST		boil-CAUS-PAST
‘boiled’		‘made boil’
- c. Lezgian (Northeast Caucasian; Haspelmath 1993, 174-5)

q^hi -jağun	vs.	ki< x >ligun
REP-hit		<REP>look
‘hit again’		‘look again’

In (37a), the exponent for the 3MS suffix is either *-h* or *-u*, depending on whether its stem ends in a vowel or consonant, respectively. In (37b), the exponent for past tense is *-i* elsewhere but *-d* after the causative marker. And in (37c), the repetitive exponent is a prefix *q^hi-* with some roots (an arbitrary

²⁵There are also cases empirically in between these two types, sometimes termed *partially* suppletive, where there’s some overlap in form between two exponents, e.g., *France* vs. *French* (Spencer 1991). We will not treat these as separate from fully suppletive allomorphy here.

²⁶The interested reader is referred to Embick and Shwayder 2018 and references therein.

list), and the infix *-x-* with other roots (a much larger list). In all these cases, the exponents are (arguably) not phonologically related to each other, and hence the different forms need to be stored, with some mechanism of the grammar deciding which will appear, in some way taking context into account.

DM, a non-lexicalist realizational theory, puts allomorphy in its own theoretical bucket, with the mechanism of Vocabulary Insertion being able to include conditioning environments, essentially as a form of subcategorization.²⁷ For example, vocabulary items along the following lines could account for the allomorphy in (37):

- (38) DM analysis of suppletive allomorphy
- a. 3SG.M $\rightarrow$ -h / C__ (Moroccan Arabic)
 3SG.M $\rightarrow$ -u / elsewhere
 - b. PAST $\rightarrow$ -d / CAUS__ (Adıyaman Kurmanji)
 PAST $\rightarrow$ -i / elsewhere
 - c. REP $\rightarrow$ q^hi- / __‘hit’ (etc.) (Lezgian)
 REP $\rightarrow$ -x- / elsewhere

The above vocabulary items essentially give the instruction that they can only be inserted in a particular context. The Elsewhere Condition (see §2.1.4) adjudicates among vocabulary items. As we will see throughout §4, DM uses the theoretical tool of contextual conditioning to handle other apparent form-meaning mismatches as well. For an overview of allomorphy from a DM perspective, the reader is referred to Gouskova and Bobaljik to appear and references therein.

The other theories that we have been discussing throughout this Element do *not* make use of a specific theoretical mechanism for suppletive allomorphy. Nanosyntax, a theory otherwise similar to DM, does not allow contextual conditioning, instead necessarily analyzing apparent allomorphs as corresponding to distinct syntactic structures (see, e.g., Caha 2024); this type of analysis stems from the theory’s treatment of syncretism and portmanteaux (see §4.3-4). MaS, a non-lexicalist but incremental approach, allows only strict one-to-one mapping between meaning/function and form, and therefore must analyze all apparently distinct allomorphs as actually distinct morphemes, not exponents of the same morpheme. The one exception in MaS is that nodes can be null in certain contexts, a kind of exponent alternation.

²⁷See Kalin and Rolle 2023 for more on the type of subcategorization involved in suppletive allomorphy.

Finally, PFM, a lexicalist realization theory that eschews pieces and morphological structure in favor of (nested applications of) processes in the context of sets of features (see §2.2), treats allomorphy as essentially a subcase of multiple exponence (see §4.2). Specifically, both allomorphy and multiple exponence are dealt with through realization rules that reference multiple features at once. For allomorphy, the set of features that are realized includes the features that DM would separate out as *context*, rather than content (see discussion in Stump 2001, 10-11).

A few robust cross-linguistic findings about suppletive allomorphy have informed the development of morphological theory, in particular, in DM and related non-lexicalist approaches. First, locality: allomorphy is triggered by the local environment and is sensitive to intervention, which (in the usual case) causes a retreat to a default (see early discussions in Aronoff 1976; Siegel 1978; Allen 1979). The locality of allomorphy is much-discussed in particular in search of a universal definition of locality. Over the years, a number of competing locality conditions have appeared (see, e.g., Bobaljik 2000; Embick 2010; Merchant 2015; Moskal 2015; Choi and Harley 2019; Paparounas 2024). What repeats across this literature on locality is that there is a role for the following factors: structure, linear adjacency, and domains. In addition, it often seems that zero exponents are able to be ignored in the conditioning environment, leading to the proposal in Embick 2010 that such nodes are “pruned”, that is, removed before exponents are chosen/inserted. Related to locality is the much-discussed absence of so-called *ABA suppletion patterns in contexts of morphosyntactic containment; see Bobaljik 2015 for an overview.

Another robust cross-linguistic finding that has received a lot of theoretical attention is that phonologically-conditioned suppletive allomorphy is generally “inward”-looking, i.e., the trigger of such an alternation is found in a more deeply embedded location relative to the alternation itself (e.g., Carstairs 1987, 1990; Bobaljik 2000; Paster 2006). In (37a)/(38a), for example, it is the phonology of the root that conditions allomorphy of the suffix, and crucially not vice-versa. On the other hand, other kinds of suppletive allomorphy can look “outward”, at less embedded material, like in (36a), where a root’s form depends on the features of an affix. These findings support proposals that are realizational, with exponent choice proceeding from the bottom up, or inside out, in order to capture inward-outward asymmetries. For discussion of this type of approach, see Kalin 2025; Kalin and Weisser To appear.

Theories that are incremental, process-based, and/or lexicalist are not always able to account for suppletive allomorphy in ways that seem motivated by the empirical landscape. For example, to the extent that structure and intervention are implicated in the locality of suppletive allomorphy (see above),

an approach that eschews structure—like PFM—cannot capture this. As another example, the possibility for outwardly-sensitive suppletive allomorphy, like that in (36a), is a challenge for incremental theories, because there should never be accessible “outward” material visible in the context. Finally, allomorph choice sometimes seems sensitive to context *across* words, in very particular configurations (see, e.g., Wolf 2008; Pak 2016; Bobaljik and Harley 2017), as well as to feature configurations arising in syntactically-derived environments (see e.g. Bonet 1994; Walkow 2012), both of which are challenging to account for in lexicalist approaches.

4.2 Multiple exponence

Instead of an abundance of exponents seeming to *compete* with each other (see §4.1 above), there may be an abundance of *co-occurring* exponents that all seem to encode a particular meaning/function. Multiple exponence—sometimes called extended exponence—has been defined as “the occurrence of multiple realizations of a single feature, bundle of features, or derivational category in more than one position in a domain” (Harris, 2016, 8).²⁸

There are a few characteristic properties of multiple exponence. Consider the complex noun in (39), meaning ‘words’ in Amharic, which has two co-occurring suffixes marking plural, *-at* and *-otftf*:

- (39) k'al-**at-otftf**
 word-PL-PL
 ‘words’ (Amharic, Kramer 2016a, 530)

In (39), plural is multiply expounded. An important factor in establishing this as multiple exponence is that both *-at* and *-otftf* independently distribute as plural markers, in other contexts (Kramer 2016a). Similarly, in the complex Amharic verb in (40), first person and plural number comprise a feature *bundle* that is multiply expounded, with *inni-* and *-än*.

²⁸Traditionally, multiple exponence is considered to be distinct from circumfixation (Harris, 2016, 19). Multiple exponence involves a single meaning/function corresponding to (or being encoded in) multiple exponents, whereas a circumfix is a single meaning/function corresponding to a single bipartite (or internally complex) exponent. One common diagnostic for circumfix status is whether one or both parts of the exponent can occur independently from each other; if they cannot, they likely comprise a circumfix. All the examples of multiple exponence in this section involve exponents that can occur independently from one another and thus they are instances of multiple exponence, not circumfixes.

multiple exponence. Under common Minimalist assumptions, it is possible for multiple probes—and the features they collect via Agree—to become part of the same word; if these multiple probes end up (independently) agreeing with the same nominal, then what looks like multiple exponence will actually just be one-to-one correspondence with multiple instances of the same feature. Thus, to establish multiple exponence, you cannot simply count exponents in a word.

A related complexity in identifying multiple exponence comes from the semantic correlates of multiple exponents. If a given (apparently) multiply-exponent feature contributes more than once to the interpretation of the word—e.g., a plural of a plural—this is not true multiple exponence. Along similar lines, it could be that what appear to be multiple exponents of one feature are in fact exponents of subtly-different features. For example, are the two plural markers in (39) actually marking the *same* plurality? Or are they marking different aspects of, or different kinds of, plurality? If the plurality is not exactly the same, this example would again not constitute a case of multiple exponence, and instead, it would be one-to-one mapping.

A final area of complexity emerges from the interpretation of variability in the form of a root/stem. Compare (39), where there were two distinct suffixes for plural, to another Amharic example, (42b), where there is one plural suffix and an accompanying change to the nominal stem (cf. (42a)).

- (42) a. wäyzäro
lady
'lady'
- b. wäyzazir-ot[tf]
lady.PL-PL
'ladies'
- (Amharic, Leslau, 1995, 175)

Unlike (39), (42b) could belong in the empirical suppletive allomorphy bucket (see §4.1), with a root exponent whose shape is conditioned by plural in the environment, and just one plural exponent, the suffix. But, in a theory that doesn't have a context vs. content distinction, like PFM, (42b) is necessarily a case of multiple exponence, with plural part of the content of both exponents (see e.g., Matthews 1972, Baerman and Corbett 2012, Caballero and Harris 2012, Harris 2016).

In terms of morphological theory, the sheer existence of (at least some true cases of) multiple exponence favors realizational rather than incremental theories. Incremental morphological theories are additive, meaning that all cases of multiple exponence must be analyzed as only apparent, i.e., as one-to-

one mappings, e.g., by semantic distinctness, by multiple underlying instances of a feature, or by contextual conditioning (see discussion above). If a theory is further constrained to disallow contextual conditioning of exponents, like MaS is, that analytic path is ruled out, and the options for analyzing apparent multiple exponence are yet further constrained. (See more extensive discussion on the problem for incremental theories in Stump 2001, p. 4-7 and Siddiqi and Boyer To appear.)

Realizational theories themselves can be differentiated from each other based on how they treat multiple exponence. The process-based theory PFM posits rules that realize one or more features (see §2.2), and since there is no constraint on how many rules in distinct rule blocks can refer to the same feature(s), multiple exponence requires no special operations at all. And indeed, as there is no contextual conditioning of exponence in PFM (see §4.1), multiple exponence abounds.

On the other hand, the realizational piece-based theory DM assumes that the correspondence between feature (bundle) and exponent is one-to-one, and *does* make a content-context distinction. The routes to what looks like multiple exponence in DM include those discussed above—multiple probes giving rise to multiple instances of a feature (bundle), semantic differentiation among instances of a feature, and contextual conditioning/allomorphy. Another mechanism sometimes utilized is a morphological operation which splits a syntactic node into two, potentially copying features as it does so, usually called Fission (for varying implementations see Noyer 1997, Trommer 1999, and Hewett To appear, i.a.).²⁹ These morphological mechanisms in DM apply in constrained ways, e.g., with contextually-conditioned exponent choice affected by locality, and with morphotactic/markedness constraints motivating node splitting.

Finally, Nanosyntax, which is realizational but eschews contextual allomorphy and all post-syntactic mechanisms, including Fission (see §2.3), is in the same boat as incremental approaches. The same analytic paths present themselves, e.g., an apparent case of multiple exponence could be reanalyzed as involving *distinct* features or as the underlying presence of multiple instances of a feature, presumably with semantic effects.

For further discussion of multiple exponence, see the classic references Caballero and Harris 2012 and Harris 2016 and the recent overviews in Fenger 2023, Puškar-Gallien et al. 2025, and Siddiqi and Boyer To appear.

²⁹See also Grofulović and Müller 2023 for another approach to multiple exponence in DM, involving feature copying without Fission.

4.3 Syncretism

Turning now to abundance on the meaning side of a form-meaning mapping, it is quite common cross-linguistically to find a particular phonological string appearing in different contexts, seeming to have *related* meaning/functions but also (to at least some extent) *distinct* meanings/functions in those different contexts. Consider the English data in (43):

- (43)
- a. She walk-**s** her two dog-**s** every day.
 - b. The girls prepared for **their** tests. (cf. for her test)
 - c. The boys prepared for **their** tests. (cf. for his test)
 - d. I walk-**ed** for 3 hours yesterday.
 - e. I have walk-**ed** for 3 hours already today.
 - f. If I walk-**ed** more often, I would drive less.
 - g. If I walk-**ed** for 3 more hours tomorrow, I would make it to the peak.

In (43a), *-s* appears as a verbal agreement marker (3rd singular, present tense) and as a nominal plural marker. In (43b-c), *their* is used as the 3rd person plural possessive pronoun irrespective of the gender of the referents (and of course can also be used for certain singular referents; see Bjorkman 2017; Conrod 2019). In (43d-g), *-ed*, typically associated with past tense, appears with a range of interpretations/functions, including as a participial marker, in the antecedent of a counterfactual, and in future less vivid environments.

The types of patterns in (43) are sometimes all characterized descriptively as syncretism. However, some of these instances of “syncretism” are accidental, where the two exponents have the same phonology but are not (at least obviously) grammatically related in any way; we will refer to such cases as ACCIDENTAL HOMOPHONY rather than syncretism. We consider (43a) to be an instance of accidental homophony given the lack of (apparent) grammatical connection between plural number and 3rd singular present tense agreement. We set aside accidental homophony henceforth, although we recognize that it is not always obvious which instances of this kind of pattern are accidental and which are not (see discussion in Harbour 2008).

Syncretism is extremely common and has been the topic of a mountain of morphosyntactic research. Prominent overviews include Baerman et al. 2005, Albright and Fuß 2012, and Caha 2019. Most syncretism research focuses on inflectional morphology; syncretisms in derivational morphology have been discussed much less, perhaps because the line between syncretism and polysemy (a form with several related meanings) can be difficult to draw, and/or because there is relatively less research in morphosemantics in general.

Syncretisms can descriptively be divided into two types, based on (at least apparent) natural class or non-natural class behavior: (i) **SHARED-FEATURE SYNCRETISMS**, where the contexts that are syncretized over exclusively share one or more linguistic features and form a natural class on that basis, and (ii) **NON-SHARED-FEATURE SYNCRETISMS**, where the contexts that are syncretized over are grammatically related but do not seem to exclusively share any features (in other words, they do not seem to form a natural class). The use of the exponent *you* for all 2nd person pronouns in English regardless of gender, and across nominative and accusative case environments (even though these distinctions are marked elsewhere in the pronominal system of English), is a shared-feature syncretism. In contrast, the inflected copula *are* displays a non-shared feature syncretism, appearing with 2nd person singular and plural subjects, as well as 1st and 3rd person plural subjects. Another sort of profile of non-shared feature syncretism is found in the use of the same exponent to mark past tense, past participle, counterfactual, and future-less-vivid meanings, (43d-g). Importantly, whether or not a syncretism is of the shared-feature type requires a particular analysis of the morphological, syntactic, and semantic context (tacitly or not), and so scholars might disagree for any particular case as to whether it is a shared-feature syncretism or a non-shared-feature syncretism, or perhaps not a syncretism at all.

In this section, we take up shared-feature syncretism (§4.3.1) and non-shared feature syncretism (§4.3.3), with a detour to metasyncretism along the way (§4.3.2). Our approach in these sections draws heavily from the recent overview paper on syncretism Kramer to appear, but with new examples and updated discussion. Then, §4.3.4 turns to cases where an exponent appears to have more significantly different meanings/functions in different grammatical contexts, as in (43d-g), often termed “repurposing”.

4.3.1 Shared-feature syncretisms

As defined above, shared-feature syncretisms are those where the syncretic exponent appears to distribute across a natural class of distinct insertion sites/environments. Syncretism thus seems to show that exponents need to be able to *compete* with each other, e.g., regulated by the Elsewhere Principle. This, in turn, is most straightforwardly implemented in a realization approach, where exponents are logically separate from contexts of insertion (see discussion in Stump 2001, Ch. 1). The main approach to shared-feature syncretisms across realizational theories is the **UNDERSPECIFICATION** of exponents: the syncretic exponent is associated only with a subset of potentially-specified features; specifically, the exponent is associated with just the shared feature(s)

of the syncretized contexts.

In DM, underspecification is represented in vocabulary items. Consider the following simplified analysis, zooming in on third person pronouns in English, *she*, *he*, *it*, and *they*, based around the features 3 (for third person), D (the syntactic category of pronouns), $\pm$ PL (for number), and $\pm$ FEM (for gender of animates), which alternates with $-$ ANIM (for inanimates). The relevant vocabulary items could look like the following (setting aside non-nominative cases and also singular uses of *they*; see Harley, 2008b for a DM analysis of all English pronouns):

- (44) Vocabulary items in a (simplified) DM analysis of 3rd person pronouns
- a. $\text{she} \leftrightarrow [3][D][+FEM][-PL]$
 - b. $\text{he} \leftrightarrow [3][D][-FEM][-PL]$
 - c. $\text{it} \leftrightarrow [3][D][-ANIM][-PL]$
 - d. $\text{they} \leftrightarrow [3][D][+PL]$

All of the pronominal exponents in (44) are fully-specified except for (44d): the syncretic exponent *they* in (44d) shares certain features with the other pronouns (third person, the D category feature, number) but it is crucially underspecified for gender/animacy. Assuming the Elsewhere Condition, *they* can be inserted at any 3rd-person pronominal node with a plural feature regardless of whether that node also has a $[+FEM]$, $[-FEM]$, or $[-ANIM]$ feature. Because the vocabulary item for *they* is the only one insertable in a plural node, it will have a natural class distribution. (Note that we could have underspecified *they* further, as just realizing the features $\{[3][D]\}$, making it instead an elsewhere syncretism; see §4.3.3.)

In PFM, which is also realizational and assumes the Elsewhere Condition, underspecification works very similarly as it does in DM. In brief, the context for the application of a realization rule can (in fact, usually is) underspecified relative to the full morphosyntactic property set of a paradigm cell. When one rule in a rule block is less specified relative to other rules in the *same* rule block, the relatively less-specified exponent is syncretic. This is the case, for example, in the realization rule in (11) in §2.2. Underspecification results in the appearance of an exponent across multiple cells in the paradigm, which will form a natural class unless a more specific form appears in one or more cell(s). (More on this below under non-shared-feature syncretisms, §4.3.3.)

In the realizational approach Nanosyntax, syncretisms are captured through a sort of *over*-specification, because of the use of the Superset Principle rather than the Subset Principle-based version of the Elsewhere Condition. As discussed in §2.3, exponents in Nanosyntax correspond to a structure, with an

exponent able to be inserted at a phrasal node just in case its corresponding structure matches that phrase’s structure exactly or contains the structure of that phrase exactly; if more than one exponent is compatible with insertion, the exponent corresponding to the smallest structure (i.e., with the least superfluous material) is chosen. Shared-feature syncretisms of the plural *they* type, (44d), are actually not wholly-straightforward to analyze in Nanosyntax, requiring a special mechanism involving an exponent whose corresponding structure contains a “pointer” to another exponent (see Wyngaerd 2018; Wyngaerd et al. 2021). The bread-and-butter of Nanosyntax ever since Caha 2009 is instead containment-type syncretisms, which can be seen as a kind of shared-feature syncretism, with an exponent being inserted for a set of structures in a containment relationship that crucially share a core set of features. This, in turn, relies on a cartographic approach to particular semantic/morphosyntactic domains, e.g., most famously within Nanosyntax, case.

Incremental theories, where morphosyntactic elements and exponents are fundamentally linked, have certain specific issues in modeling syncretism, at least when implemented within a Minimalist (projection-based) approach to syntax as MaS is. The issue, at its core, is that an exponent that corresponds to some underspecified morphosyntactic feature(s) should in fact *not bear* the syncretized feature(s) at all. This, in turn, should constrain the morphosyntactic behavior of that exponent. If gender is syncretized in a pronoun, for example, then when that pronoun agrees with a verb, gender should also be syncretized there, because the pronoun actually *lacks* gender. This is very often not the case, empirically. (Note that unification-based approaches to syntax, like HPSG, do not have this problem.)

4.3.2 Metasyncretisms

Some syncretisms hold across a whole morphological system, applying to sets of exponents; these syncretisms have been referred to as METASYNCRETISMS (see e.g., Williams 1994, Bobaljik 2002, Harley 2008b). Consider the third person pronouns in Amharic, (45).

- (45) 3rd Person Pronouns in Amharic (Leslau 1995)
- a. *issu* ‘he’
 - b. *iss^wa* ‘she’
 - c. *innässu* ‘they’

The plural pronoun in (45c) is used for both genders, so it is tempting to just provide an underspecification analysis where the exponent *innässu* would be

underspecified for gender, like *they* in (44). However, in Amharic, *none* of the elements in the language that distinguish gender in the singular do so in the plural: the subject agreement affixes, the object marker, the possessive pronouns, the demonstratives, and so forth all have a single plural exponent used for both genders, while distinguishing gender in the singular (Kramer 2016b). Therefore, an analysis that simply treats each of these plural agreement exponents as underspecified for gender features would miss a significant generalization about the morphology of Amharic. Indeed, such an analysis would also miss a significant generalization about the morphology of all the world’s languages: gender is much more likely to be distinguished in the singular rather than the plural (Greenberg, 1966, 95).

Many morphological theories therefore treat metasyncretism in a different way than exponent-specific syncretism. For example, DM uses IMPOVERISHMENT (see e.g., Bobaljik 2002, Harley 2008b, Keine and Müller To appear), a post-syntactic operation that deletes features prior to Vocabulary Insertion. Impoverishment is helpful in accounting for metasyncretisms because it can delete specific features (e.g., gender) in a particular context (e.g., plural) such that *all* feature bundles with those features are affected. Then, when Vocabulary Insertion occurs, it will not be possible to insert an exponent that is associated with the deleted features; effectively, Impoverishment ends up constraining possible exponents in a language. The Impoverishment operation for the gender metasyncretism in Amharic deletes gender features in the context of plural features, along the lines of (46).

- (46) **Gender Impoverishment in Amharic** (Kramer to appear)
 $[+PL], [+/-FEM] \rightarrow [+PL]$

Connecting back to the cross-linguistic picture, it seems likely that (46) is motivated on the grounds of universal markedness considerations which disprefer the expression of gender and number together (see Noyer 1992).

In PFM, a RULE OF REFERRAL—a kind of identity statement between cells of a paradigm—is used to state metasyncretisms directly (see e.g., Zwicky 1985, Stump 2001, Albright and Fuß 2012; see also “property mappings” in Stump 2016). One possible statement of a Rule of Referral that would capture the Amharic gender metasyncretism is in (47).

- (47) **Rule of Referral for Gender Metasyncretism in Amharic:**
 In the plural, masculine and feminine gender exponents are identical.

Rules of referral simply state outright an identity relation, and can thus encode all kinds of different identity relations between exponents within a paradigm.

Impoverishment is more restrictive than rules of referral in that it (conventionally) only deletes features and does not change or add them. Research is ongoing into whether the predictions of a more restrictive approach to metasyncretisms is borne out; for a negative answer, see Spencer 2019 and for a more positive answer, see Kramer 2016b and Kramer to appear.

Other theories lack a mechanism dedicated to metasyncretism. In Nanosyntax, although it is realizational like DM and PFM, the rejection of all post-syntactic mechanisms (apart from late insertion of exponents) makes metasyncretism impossible to account for directly. Metasyncretisms must instead be accidental. In an incremental theory, too, metasyncretism must be accidental, or must reduce to something else; on metasyncretism and MaS specifically we refer the reader to Collins and Ordoñez 2021 and Myler 2024a.

4.3.3 Non-shared-feature syncretisms

Some syncretisms (both metasyncretisms and exponent-specific syncretisms) occur across contexts that do not exclusively share features.³⁰ In practice, this can refer to a number of different types of empirical cases. We review two examples here: (i) non-shared-feature syncretisms as defaults (elsewhere distributions of an exponent), and (ii) (apparent) non-shared feature syncretisms as (surprising yet) identical meanings/functions.

One way to get a non-shared feature syncretism is through an elsewhere distribution of an exponent, where there are specific exponents for some cells of a paradigm and then a default form appears in the rest of the cells. Underspecification (e.g., for PFM and DM) is again an effective tool for analyzing such cases. A simple example of this is furnished by the English present tense copula, which can have three forms, *am*, *is*, *are*. One analysis of these forms takes *am* to have 1SG features, *is* to have 3SG features, and *are* to appear for all other persons and numbers, as the default (an elsewhere distribution, arising due to the absence of features). Elsewheres/defaults thus are a sort of concealed shared-feature syncretism, just one that doesn't end up covering a neat natural class. Note that it is not always obvious that a certain form is the default; see, e.g., Kramer and Teferra 2020 on Sidaama (Cushitic), where feminine corresponds to the default (underspecified) exponent for gender.

³⁰This type of syncretism is sometimes referred to as a MORPHOME—an element with a purely morphological distribution, with no grounding in syntax or semantics—especially when it is a metasyncretism. The term “morphome” was introduced in Aronoff 1994, where it was used to characterize a non-shared-feature metasyncretism in Latin where the same stem allomorph is found in an array of diverse grammatical contexts. See Trommer 2016, Maiden 2021, and Myler 2025 for recent discussions of morphomes.

Tunisian Arabic (Semitic) furnishes a different type of syncretism, as certain plural nouns can trigger agreement on the verb that looks the same as feminine singular agreement, as seen in (48) with the plural noun ‘men’ (agreement bolded).

- (48) el rʒe:l xerʒ-**et**
 the man.PL went.out-3PL?/3SG.F?
 ‘The men went out.’ (Tunisian Arabic; Dali and Mathieu, 2021, 79)

This is not analyzable as a default because masculine is the default gender in Tunisian Arabic (Dali, 2020, 40-41), and therefore an underspecification analysis is not appropriate. The use of feminine singular agreement markers for 3rd plural nouns only occurs with nouns that have a certain kind of plural inflection (the non-concatenative “broken plural”).

Non-default non-shared-feature syncretisms, like that described for Tunisian Arabic above, are straightforward to account for in some approaches to morphology, but challenging or impossible in others, requiring a different kind of analysis. In PFM, rules of referral can be straightforwardly formulated to capture such cases. For DM, Nanosyntax, and MaS, the most straightforward account would be to analyze agreement in (47) as *actually* being 3FSG agreement, which means there must be an independent source for feminine singular features in the morphosyntactic structure. If this is on the right track, we expect that plural nouns which trigger feminine singular agreement in Tunisian Arabic should have different properties or behave differently than other plural nouns. This is actually borne out in several respects, as described extensively in Dali and Mathieu 2021. Indeed, there is a substantial literature on how non-shared feature syncretisms are often better analyzed as either not syncretisms at all or as shared-feature syncretisms, once the case has been better understood (see e.g., Kalin and Atlamaz 2018, Francez and Koontz-Garboden 2016, Kramer to appear and the works referred to therein). As we will see, this is true, too, for the more radical syncretisms discussed next.

4.3.4 Repurposing as syncretism

REPURPOSING refers to cases where an exponent appears to have *significantly* different meanings/functions in different contexts, often characterized as having a core function and a secondary or repurposed function. These are, essentially, apparent non-shared-feature syncretisms, some of which have been well-studied from a semantic perspective.

Recall from (43) the wide distribution of English *-ed*, in particular as marking both past tense and counterfactuality. If we investigate more closely, we

would see that it's not just the specific *-ed* past exponent that marks counterfactuality: all irregularities of past marking also appear in (non-past) counterfactual environments. This particular phenomenon is also not unique to English: in (49) the Zulu (Bantu) sentences and their English translations both contain past tense morphology (bolded) in past and present counterfactuals (CFs) as well as future less-vivid (FLV) environments, even when it conflicts with the temporal interpretation of the clause:

- (49) Counterfactuals in Zulu (Halpert and Karawani, 2012, (9))
- a. [ukuba **be**-ngi-phum-e izolo]
if PST.IPFV-1SG-leave-PFV yesterday
be-ngi-**zo**-fik-ile ekuseni
PST.IPFV-1SG-FUT-arrive-PFV LOC.dawn
'If I **had left** yesterday, I **would have** arrived at dawn.'
(past CF)
- b. [ukuba **be**-ngi-phuma manje] **be**-ngi-**zo**-fika
if PST.IPFV-1SG-leave now PST.IPFV-1SG-FUT-arrive
kusasa
tomorrow
'If I **left** now, I **would** arrive tomorrow.'
(present CF)
- c. [ukuba **be**-ngi-**zo**-phuma kusasa]
if PST.IPFV-1SG-FUT-leave tomorrow
be-ngi-**zo**-fika ngo-Lwesihlanu
PST.IPFV-1SG-FUT-arrive on-Friday
'If I **left** tomorrow, I **would** arrive on Friday.'
(FLV)

Are such cases instances of accidental homophony, or some type of syncretism, or actually just identicalness? This particular kind of pattern—where past tense morphology also marks counterfactuality—occurs in a wide variety of typologically unrelated languages, suggesting that the connection between the two is not accidental (e.g. Iatridou, 2000; Arregui, 2009; Karawani, 2014). Iatridou (2000) characterizes the appearance of past tense (and other temporal) morphology in counterfactuals as “fake”, since it does not receive a temporal interpretation, which she assumes to be its primary function. Beyond counterfactuals and past tense, we find many parallel types of patterns throughout the world’s languages (e.g., Izvorski 1997; Bjorkman and Cowper 2016).

As with the other non-shared-feature syncretisms discussed in §4.3.3, there are different approaches to analysing repurposing. One possibility is to treat repurposing as an elsewhere syncretism, with the relevant morphology as an unexpected default: for example, certain temporal morphology could be un-

derspecified (e.g. for tense and/or aspect), so we can get the illusion of fixed temporal settings that are really just a default realization of a verb or auxiliary (e.g. Bjorkman and Halpert, 2011, 2017; Halpert and Karawani, 2012; Karawani and Halpert, to appear). Another possibility is to treat repurposing as involving a shared feature, with this feature being highly abstract and shared across diverse environments (e.g., Iatridou 2000’s exclusion feature). A final possibility is that the exponed feature/feature set could actually be *identical* across different environments, and highly-specified—making this not a syncretism at all—with the feature(s) having different effects in the mapping to LF depending on the environment (e.g. Arregui, 2004).

In doing the kind of careful work needed to understand cases of repurposing, we are often led to deep and surprising morphosyntactic and semantic similarities that otherwise would have eluded us.

4.4 Portmanteaux

A portmanteau (plural: portmanteaux or portmanteaus) is a single exponent that appears to correspond to multiple morphosyntactic units of some kind (e.g., multiple features, multiple morphological “slots”, multiple syntactic terminal nodes, etc.). To start with a much-discussed example, in French (Romance), the preposition *de* ‘of’, which is usually its own word (see (50a,b)), forms a unit with a singular definite masculine article when the following noun is consonant-initial, as in (50c) (cf. (50d)).

- (50)
- | | | | |
|----|---------------------------|--------|----------|
| a. | de la | mère | |
| | of DEF.F.SG | mother | |
| | ‘of the mother’ | | |
| b. | de l’ | ami | |
| | of DEF.M.SG | friend | |
| | ‘of the friend’ | | |
| c. | du | père | |
| | of.DEF.M.SG | father | |
| | ‘of the father’ | | |
| d. | *de le | père | |
| | of DEF.M.SG | father | |
| | Intended: ‘of the father’ | | (French) |

The phonological string *du*—assuming it is a single exponent (more on this below)—is a portmanteau because it simultaneously encodes the preposition and the definite article.

Portmanteaux are also found in the clausal domain. For example, many languages in the Macro-Sudan belt (Güldemann, 2008) combine the phi-features of the subject with some combination of tense, aspect, mood and polarity; they are accordingly called STAMP morphs and are clear examples of portmanteaux (see e.g., Anderson 2016, Garvin et al. To appear for recent typological overviews). In Tarok (Benue-Congo), there is a STAMP portmanteau that combines the person/number of the subject with aspect, as in (51a), and one that combines the person/number of the subject with mood, as in (51b), among others.

- (51) a. **n** wá ú-diŋ
 1SG.PFV drink CLF-water
 ‘I have drunk the water.’
 b. **mi** wá a-tí ipín
 1SG.IRR drink CLF-tea tomorrow
 ‘I will drink tea tomorrow.’
 (Tarok; Anderson 2016, 514, Sibomana 1981, 238)

Again assuming the bolded phonological strings are not further decomposable, these are portmanteaux because they simultaneously encode distinct grammatical categories.

When defined as the conglomeration of distinct morphosyntactic categories, portmanteaux are found very commonly cross-linguistically. Specifically, portmanteaux are common with respect to *inflectional* categories, like tense, aspect, mood, and agreement in the verbal domain, and case, number, and gender in the nominal domain. Indeed, on the basis of having many such portmanteaux inflectional forms, some languages are termed FUSIONAL in the typological literature.

From a morphology/syntax interface perspective, there are a few key issues in identifying portmanteaux. As a first consideration, can the portmanteau be decomposed, perhaps counter to initial appearances? One could argue that the portmanteaux that we have seen so far are not further decomposable into separable exponents. Importantly, though, this is a matter of argument and/or analysis. For example, see Newell et al. 2024 for a compositional analysis of the French portmanteau in (50c). STAMP morphs are also at least sometimes decomposable, for example, in Lobi (Mabia/Gur), where they can be separated into a subject agreement exponent and a TAM exponent (Garvin et al., To appear), modulo predictable phonological alternations.

Beyond decomposability, determining what is a portmanteau and what isn’t depends crucially on theory-specific assumptions about what constitutes

the relevant morphosyntactic unit, naturally differing based on lexicalism vs. non-lexicalism. Non-lexicalist approaches, given their basis in syntax, generally reserve the term “portmanteau” for cases where separate syntactic heads are realized with one exponent. In turn, the treatment of such multi-head portmanteaux depends further on whether a non-lexicalist theory is cartographic or not, and realizational or not.

Cartographic, non-lexicalist approaches hold that all morphosyntactic features are their own heads in the Syntax. Cartography combined with late insertion at phrasal levels, as found in Nanosyntax, means that *any* bundling of features expressed by one exponent necessarily implicates a portmanteau exponent, and portmanteaux abound. Cartography combined with early (incremental) insertion at the terminal node level, as found in MaS, raises the issue that all apparent portmanteau must be introduced by a single (non-featurally-complex) head; further, all other heads that seem implicated in the portmanteau in terms of meaning/function must in turn be silenced (see, e.g., Myler 2024a). This challenge is similar to that faced by any incremental approach (lexicalist or non-lexicalist), in that there is potentially a lookahead issue: if there are other exponents that express subparts of the portmanteau in other contexts, these need to be blocked in lieu of the portmanteau in just the right configuration of features.

Non-cartographic, non-lexicalist approaches like DM allow for syntactic heads to consist of bundles of features. In this context, a bundle of features being realized by one exponent is a true portmanteau only if the bundled features originated in different syntactic heads; as such, theories that allow pre-syntactic feature bundling allow what is and isn’t a portmanteau to vary by language. For portmanteaux that don’t seem to be due to pre-syntactic bundling, four analytic routes have emerged in the literature.³¹ First, especially in cases involving agreement, it may be that features from different morphosyntactic loci are collected on one terminal; see, e.g., Georgi 2013. Second, some apparent portmanteaux have been analyzed as an alternation on just one head, conditioned by the presence of other specific features on other heads; in other words, a portmanteau might actually just be displaying suppletive allomorphy (§4.1); see Trommer 2010, Fenger 2018, Fenger 2023, and Banerjee 2021. Third, the post-syntactic operation Fusion has been invoked to *create* a single unified node from distinct syntactic nodes; a single exponent can be then be inserted at the new node (see e.g., Noyer 1992, Halle and Marantz 1993). Finally, some approaches have argued for the possibility

³¹See discussion in Šereikaitė 2025 on how to tease apart pre-syntactic bundling of features from syntactic and post-syntactic bundling.

of *directly* realizing a series of syntactic nodes with a single exponent. This is true of Ostrove’s (2018) Stretching, based on linear adjacency, Svenonius’s (2012) Spanning, based on a complement sequence (see §3.3.2), and Radkevich’s (2010) Vocabulary Insertion Principle, which allows limited insertion at non-terminals. Such proposals are motivated by the observation that portmanteau formation often seems to be syntactically-conditioned or restricted, to immediately local nodes.

A different kind of approach to portmanteaux is found in lexicalist, realizational, process-based theories like PFM. In such theories, words do not have hierarchical structures, rather, exponence is determined by ordered rule blocks (§2.2). In the usual case, one rule applies per rule block, thereby exponing one “position” in the word. As discussed in §4.1-2, a realization rule can refer to more than one feature, and so there is nothing exceptional about that kind of realization rule. True portmanteaux, for PFM, are exponents that span more than one rule block, occupying what are usually two (or more) distinct positions for exponence. In this case, a separate rule block is used for the insertion of a portmanteau exponent; this rule block prevents other specific rule blocks from applying (see e.g., Stump, 2001, 139-144, Stump, 2016, 235-237). To the extent that portmanteau formation is syntactically-conditioned or restricted, this of course cannot be captured in such a system, though linear conditioning could be.

There are few cross-theoretic or even typological overviews of portmanteaux, perhaps because the empirical scope of the phenomenon depends more directly on theoretical assumptions. However, see Radkevich 2010, Ch. 4 for a typological survey of TAM portmanteaux and a comparison of different theoretical approaches.

4.5 On absence in mappings

The final two form-meaning mappings we turn to, briefly, are those that involve *absence* on one side of the mapping: absence of *form*, i.e., morphological gaps, §4.5.1, or absence of *meaning/function*, i.e., empty morphs, §4.5.2. Both phenomena pose certain challenges for essentially all morphological theories.

A preliminary note here is that another kind of absence of form is null exponence, e.g., the plural of *fish* in English is still *fish*. There are debates about how to deal with null exponence, though the differences that emerge among approaches largely come down to theory-internal assumptions: in a process-based theory like PFM, null exponence is attributed to the identity function; in a piece-based theory like DM, null exponence is attributed to an exponent that happens to be null (lack phonological features). Null exponence has

sometimes also been taken to indicate the *absence* of a morpheme/feature altogether, most commonly posited for instances of crosslinguistically unmarked features, like nominative case (see, e.g., Aissen 1999). For space reasons, we will not further explore null exponence here.

4.5.1 Morphological gaps

A morphological gap arises when a predicted morphological form is not attested, and speakers do not uniformly or consistently accept any potential word form in its place. Some such gaps are **SYSTEMATIC**, and exist for all relevant forms in a language (e.g., all verbs exhibit the gap). Many systematic gaps have a plausible syntactico-semantic explanation. In Hixkaryana (Carib), for example, verbs agree with both subject and object, but no agreement forms exist for subject-object combinations with overlapping reference (e.g. “I saw myself” or “we (incl.) saw you”) for any verb (Derbyshire, 1985, 188-190). But other gaps appear to be idiosyncratic to particular lexical items, often called **ARBITRARY** gaps (see e.g. Hetzron 1975). The literature on morphological gaps has been mostly concerned with this latter type, as idiosyncratic gaps pose a greater challenge for linguistic theories.

Arbitrary morphological gaps are not uncommon cross-linguistically. In English, for example, the irregular verb *stride* lacks a past participial form for many speakers; some express the intuition that the past participle “should be” *stridden*, but nonetheless reject that form. Similarly in Russian (Slavic) there are verbal paradigms that are considered “defective” because they include lexically-specific arbitrary gaps. For example, the verb *pylesosit* ‘to vacuum’ lacks a first-person singular present tense form; as Pertsova 2016 notes, while speakers can predict a phonologically well-formed first-person singular form, they do not accept it. Cases where a particular inflected form simply has no licit realization have been called **INEFFABLE** (Pesetsky, 1997).

Why do arbitrary gaps arise? Albright 2006 suggests that arbitrary morphological gaps emerge for comparatively low-frequency items, when speakers lack sufficient evidence to decide between multiple equally plausible forms. The Tolerance Principle articulated in Yang 2016 proposes that the evidence needed for a child to adopt a regular rule increases logarithmically with the number of cases where the rule could apply. The Tolerance Principle is applied, for example, to account for variation between varieties that lack a contracted negative form *amn’t* and those that allow it (Thoms et al., 2025). Other arbitrary morphological gaps may have a phonological basis. See Fanselow and Féry (2003) for a review of cases of apparently phonologically-motivated ineffability, and potential approaches within Optimality Theory.

Truly arbitrary morphological gaps pose a puzzle for many generative theories of word formation. DM, for example, lacks any mechanism for filtering some outputs as unrealizable. As Trommer 1999 notes, however, it is possible to imagine a version of DM where ineffability could result if no realization rule could apply in the specific case of a given form, that is, if there is not always a default rule of zero exponence. But, one might expect in that case for a (non-zero) default realization to apply when available, yielding for example *strided* as the past participle of *stride/strode*. Similar issues in explaining morphological gaps arise for all realizational theories, including Nanosyntax and PFM.

Sims (2015), assuming a realizational lexicalist framework, argues that far from being a fringe phenomenon, gaps are a natural part of paradigms, with gaps themselves being morphological objects of a sort. Specifically, she proposes that morphological gaps are “essentially allomorphs of abstract morphemes (i.e., sets of morphosyntactic values) and in this sense are systemic variants of normal inflectional structure” (Sims, 2015, 253). This approach may also be adaptable to other realizational theories, though to our knowledge this has not been explored.

Some morphological gaps—whether arbitrary or systematic—seem to be filled syntactically via periphrasis. The alternation in English between morphological comparatives (*fancier*) and syntactic comparatives (*more opulent*) is often described in these terms: when an adjective lacks the *-er* comparative entirely, or the syntactic context does not allow that form for whatever reason, the gap is filled by the syntactic comparative with *more*. One intuition pursued by many authors working on synthetic-periphrastic alternations is that these constitute a form of BLOCKING, an idea first formulated by Poser (1992), that the existence of a morphological way of expressing certain information (e.g. *fancier*) blocks the use of the syntactic paraphrase (*more fancy*).

Alternations between synthetic and periphrastic expressions of (what appears to be) the same grammatical information have been of interest to theories of the morphology/syntax interface precisely because they appear to show a direct interaction between morphological and syntactic mechanisms (cf. §2.1.1-2). This has been taken as evidence for a syntax-centric approach by proponents of DM, including Bobaljik 2012 for comparatives and Bjorkman 2011 for auxiliary constructions. In lexicalist approaches, this has been addressed by allowing the Morphology to generate syntactically complex expressions (e.g., Ackerman and Stump 2004).

We refer the interested reader to the discussion and further references in Kiparsky 2005 and Sims 2015.

4.5.2 Empty morphs

Another puzzling type of morphological pattern involves apparently empty morphs, a term first introduced by Hockett (1947) for stable pieces of form that appear to lack any corresponding *meaning or function*.³² Empty morphs seem to occur to meet phonological and/or morphological well-formedness requirements, and pose challenges for both lexicalist and non-lexicalist theories. A criterion for considering something to be an empty morph is that it consists of something other than likely epenthetic segments, like [ə] or [ʔ].

It is common for there to be an apparent phonological or prosodic motivation for the appearance of an empty morph. In Quechua (Quechuan), for example, the apparently empty morph *ni* occurs between nouns and possessive suffixes when the noun ends in a consonant, (52a), but not when it ends in a vowel, (52b):

(52) Cuzco Quechua (Corbett, 1994, 176)

- a. yawar-**ni**-y
blood-??-1SG
'my blood'
- b. wasi-y
house-1SG
'my house'

Corbett (1994) suggests the appearance of the empty morph *ni* may have arisen historically as a way to avoid illicit consonant clusters, though Myler (2024b) notes that *ni* occurs in at least one variety of Quechua (Santiago del Estero Quechua) that lacks any relevant phonotactic constraint. In some cases empty morphs seem to be the same as an existing, related exponent, but appearing when not morphosyntactically-motivated.

There are also myriad examples of empty morphs appearing for purely morphological reasons. Theme vowels in Romance languages seem to fit the bill (bolded):

(53) Catalan (Romance; Oltra Massuet 1999, 5-6)

- a. Class I: cant-**a**-nt 'singing'
- b. Class II: tem-**e**-nt 'fearing'
- c. Class IIIa: un-**i**-nt 'joining'
- d. Class IIIb: surt-**i**-nt 'going out'

³²Some terms that appear in the literature that cover at least some of the same empirical territory as empty morphs includes "dummy morph(eme)s", "epenthetic morph(eme)s", "stabilizers", "extenders", "interfixes", and "augments".

The theme vowel is an obligatory vowel that appears between the verb root and progressive *-nt*, but that vowel does not (apparently) bear any meaning or function itself, and varies by class of verb root. Other conjugation-class-related morphology—in particular when it is isolated from other contentful inflectional marking (e.g., tense/aspect/mood)—can also be taken to constitute an empty morph. Linkers in compounds (see (21) in §3.1.3) may also be considered empty morphs (see Szczepaniak 2020 for an overview).

Empty morphs pose a significant challenge for non-lexicalist models of morphology because such models are piece-based and syntax-first, thus in the usual case necessarily attributing some particular morphosyntactic source for all exponents. The challenge is most significant for purely syntactic approaches, like MaS, and post-syntactic approaches that have late insertion but no other post-syntactic mechanisms, like Nanosyntax. For these theories, there is no analytical option except to attribute a meaning/function to empty morphs, i.e., to analyze the empty morph as not actually empty. See, for example, Cortiula 2025 on a Nanosyntactic approach to Friulian (Romance) verbs. In post-syntactic approaches like DM, empty morphs can be handled through insertion of a morph(eme) post-syntactically that has no origin in the Syntax, sometimes referred to as Dissociated Node Insertion or node-sprouting (e.g., Choi and Harley 2019). All these theories, though—even those with a rich post-syntax—can have difficulties modeling apparently phonologically-optimizing empty morphs, depending on the posited nature of the morphology/phonology interface (see, e.g., Wolf 2008). They can also have difficulties capturing the recruitment of related morphology in cases of empty morphs, unless such morphology can be analyzed as a default of some kind (see §4.3.3).

Empty morphs can also pose challenges for lexicalist theories. In PFM, stems can have an arbitrary index assigning them to a “morphomic” (purely morphological) class (see fn. 30), which can in turn be utilized by a realization rule and result in the appearance of an empty morph. Similarly, Rules of Referral (see §4.3) can be leveraged to explain the recruiting of related morphology. Morphologically-motivated empty morphs have thus sometimes been cited as evidence for a separate morphological component, as found in PFM. Phonologically-motivated empty morphs, however, are still a challenge to account for.

4.6 Section summary

In this section, we’ve shown that the tools of different theories constrain the analysis of form-meaning mappings and make different predictions about what sorts of mappings should be found in human languages. Table 4 summarizes

	PFM	MaS	Nanosyntax	DM
One-to-one	yes	yes	yes	yes
Allomorphy (§4.1)	no	no	no	yes
Multiple exponence (§4.2)	yes	no	no	no; extra
Syncrretism (§4.3)	yes; extra	no	yes; extra	yes
Portmanteaux (§4.4)	yes; extra	no	yes	yes; no; extra
Morphological gaps (§4.5.1)	no	no	no	no
Empty morphs (§4.5.2)	yes; no	no	no	no; extra

Table 4: Form-meaning mappings and morphological theories

our discussion of four specific representative theories. Each row picks out a different type of form-meaning mapping—crucially, it is the *descriptive* buckets that are intended here (see discussion at outset of §4).

Each cell in this table then offers an answer to the question, “does this descriptive bucket receive straightforward treatment within this theory?” A “yes” value in a cell means this descriptive bucket can be attributed to true non-one-to-one mappings at some analytic level in the theory, using the most basic mechanisms of the theory. A “no” value means cases in this descriptive bucket must be analyzed as (actually) being some other kind of mapping, or as otherwise having a distinct explanation. Finally, “extra” indicates that the theory can handle that descriptive bucket with the addition of an extra mechanism that goes beyond the basic theory. Multiple values in a cell indicate that the answer is different for different subcases of the descriptive bucket.

As seen in Table 4, the purely-syntactic approach MaS effectively denies there are any true non-one-to-one mappings, necessitating reanalysis of all such apparent cases (a tall order!). Most significantly opposed to MaS are the theories PFM and DM, which are alike in positing a realizational morphology *and* the existence of mechanisms in the Morphology beyond just realization. In between these two extremes is Nanosyntax, which is realizational but does not generally permit other morphological mechanisms. A bird’s-eye view of theories and mappings shows clearly that simplification to the theory comes with tradeoffs in empirical coverage, making form-meaning mappings a crucial area of research for the morphology/syntax interface.

5 Conclusion

In this Element, we have covered a lot of ground, both empirically and theoretically, in the rich and ever-growing scholarship on the relationship between morphology and syntax. Foremost in understanding this relationship is the question of whether syntax and morphology interface at the level of the word only, and relatedly, whether there is a computational system that deals in the principles and processes involved in words and word-formation—the Morphology—that is separate from the computational system dedicated to linguistic expressions larger than a word—the Syntax. As discussed extensively in §2, theories run the whole gamut of possibilities. This makes for a rather complex theoretical landscape to navigate for a scholar interested in doing research at this interface, a landscape we have aimed to illuminate, alongside what we take to be core empirical considerations, namely, wordhood (§3) and the nature of form-meaning mapping (§4). Of course, in any work of constrained size, there are many empirical and theoretical areas that have to be put to the side, such as (here) perspectives from morphosemantics, acquisition, diachrony, and language processing. These different perspectives have the potential to bring to light different sorts of explanations for why morphological phenomena look the way they do.

A central goal of this Element was to show that different theories privilege different considerations, all best capturing particular slices of, and/or perspectives on, the empirical picture. The assumptions and tools of a given theory inevitably direct the investigator’s gaze and constrain analyses: lexicalist approaches compel researchers to see the *differences* between morphology and syntax and the ways in which the unit of “word” defines a unique domain in grammar, while non-lexicalist approaches bring into focus the *similarities* between morphology and syntax and the parallels across linguistic expressions of all sizes. Piece-based approaches require decomposition of all linguistic expressions into discrete concatenating units (even when this is far from straightforward), while process-based approaches accept essentially any form of exponence as possible. And realizational approaches trivially handle (at least some) mismatches between feature/meaning and phonological form, while such apparent mismatches often compel incremental approaches to search for alternative analyses, e.g., unexpected morphosyntactic elements and distributions. Genuine discoveries that push forward our understanding of human language have been born from all these different starting points, since each perspective motivates deeper investigation of different sorts of phenomena.

While there is a remarkable lack of consensus in the study of the morphology/syntax interface, there is nevertheless a significant amount of convergence

around one or the other (or both) of the following hypotheses: (i) that morphology is realizational in nature, i.e., phonological exponents are introduced separately from the features/meanings they correspond to, and (ii) that morphology is (largely or entirely) governed by the the same computational system as syntax, i.e., there is not a wholly syntactically-independent morphological module. All the influential theories that we’ve been discussing throughout this Element adopt at least one of these: PFM is based around realizationality but rejects the unity of morphology and syntax; MaS is the reverse, rejecting realizationality but equating morphology and syntax; and DM and Nanosyntax embrace both proposals, though each in quite different ways.

Many linguists encounter only one type of approach to morphology and take for granted that the backbone of the theory they’ve been taught is orienting their gaze “correctly”. But especially in the study of the morphology/syntax interface—where there are so many parameters of cross-theory variation, and the data are so rich and multi-faceted—it is crucial, for the progress of the field, to take an aerial view. In doing so, we can let the data speak in a way it cannot otherwise, and ultimately, this will enable us to discover more illuminating cross-linguistic generalizations and make stronger theoretical arguments. We hope this Element serves as a stepping stone for scholars interested in doing research at the morphology/syntax interface, and contributes to more linguists embracing the aerial view we have aimed to offer.

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